THE PRACTITIONER.

MAY, 1869.

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*** START OF THE PROJECT GUTENBERG EBOOK THE PRACTITIONER. MAY, 1869. ***

THE PRACTITIONER.

MAY, 1869.

Original Communications.

ON THE RESTORATIVE TREATMENT OF PNEUMONIA.

BY JOHN HUGHES BENNETT, M.D., F.R.S.E.

Professor of the Institutes of Medicine, and Senior Professor of Clinical Medicine in the University of Edinburgh.

I have long formed the opinion that the prevailing method of determining the value of any particular medicine or mode of treatment is essentially faulty. Practitioners, after watching a few cases, form a favourable opinion of this or that mode of procedure; they then publish their views, supporting them with their successful cases, and strongly recommend them to the consideration of their medical brethren. Then follow trials more or less numerous by others, some of whom think the method recommended good, whilst others find it useless or injurious. Such a system is characteristic of an imperfect acquaintance with medicine, and during the progress of many centuries, while it has led to some valuable knowledge, has for the most part only tended to superficiality and the utmost contrariety in medical practice. What seems to be necessary at present for determining the real value of any kind of treatment is—

1st. Rigid accuracy in the diagnosis of the case.

2d. A clear comprehension of the nature of the pathological condition treated.

3d. An acquaintance with the natural progress of the disease.

And

4th. A tabulated account of the cases treated, showing the care with which they were observed, and their chief symptoms, including the time they were under treatment, and the termination in success or failure.

Doubtless this method of determining the value of any treatment requires a high degree of medical knowledge, and some trouble; but I would suggest that it is the only one capable of inspiring confidence and permanently advancing the interests of the medical art. If it cannot be carried out during the exigencies of every-day practice, there is nothing to prevent its prosecution in our public hospitals, where the patients are under constant observation, and where there are in many of them a staff of assistants whose business it is to make the necessary records.

The chief obstacle to obtaining accuracy in result is the general conviction among medical practitioners that a different treatment is required, even in fixed morbid conditions, according to the symptoms which may be present. The progress of diseases is never absolutely uniform, and no doubt the occurrence of particular phenomena often require special interference. This secondary treatment of symptoms, however, should never be allowed to interfere with the primary management of the morbid condition; and it is the neglect of this rule which has led to such injurious results in the treatment of many diseases. If, for example, in order to relieve cough in phthisis we give opiates and expectorants, how can we maintain the appetite and improve the tone and digestibility of the stomach, on which the assimilation of food, cod-liver oil, and nutrition essentially depend?

Since the publication of my papers and treatise on the Restorative Treatment of Pneumonia I have watched with great interest what has been published by the profession on this subject. The only published series of cases that I am acquainted with is given by Dr. T. N. Borland, of the Boston City Hospital, U.S. He tabulates according to the form I recommended 90 cases of pneumonia, of which he says twelve died—a mortality of one in 7½ cases.[1] Of these, four had phthisis; two were chronic, having been admitted on the eighteenth and twenty-first days of the disease; one was utterly prostrated on admission, and died the following day; one was a case of surgical injury, transferred to the medical wards on the occurrence of fatal pneumonia; and one was a case of typhoid fever—leaving only three fatal cases of true primary acute pneumonia. Of these, one died of cerebrospinal meningitis; a second suddenly, from supposed embolism; and a third, from extensive double pneumonia, with violent delirium. Details of the post-mortem appearances are much desired in these fatal cases. A rigid scrutiny into the true character of these cases therefore shows, instead of a mortality of one in 7½ cases, as is alleged, a real mortality of only one in 27 cases—that is, three deaths in 82 cases.

Since I published the accounts of 129 cases, on which my statistics were founded,[2] with four deaths, and a mortality therefore of one in 32¼ cases, I have treated in the clinical wards of the Royal Infirmary 24 other cases, with one death. This increases the mortality to 1 in 30⅗, in the total of 153 cases. Of these a tabulated account will be published, without which I venture to say little information can be obtained with regard to the results of any kind of treatment. Of this the analysis of the Boston cases offers sufficient proof; for although Dr. Borland says: "The greater proportion of these cases have been treated according to the plan set forth by Dr. Bennett, by restoratives directed to further the natural progress of the disease," he does not appear to have remarked that all my cases were those of acute primary pneumonia, and not consecutive or secondary cases in individuals weakened by phthisis, broken down by long starvation and surgical injuries, or such as have become chronic with gangrenous abscesses.

Dr. Popham of Cork[3] tells us that he treated 30 cases of pneumonia by the restorative plan, and that, with the exception of two who were admitted in a dying state, all recovered. In 28 cases, therefore, admitting of treatment, all recovered. It is much to be regretted that these cases were not tabulated, so that the reader might judge of their extent, severity, and progress. We are told, however, that six were cases of double pneumonia; in eight the left lung only was engaged, and the right lung in fourteen. Dr. Popham also tells us that instances occurred so grave that he did not consider himself justified in trusting to restoratives alone. He therefore gave 5 grs. of bicarbonate of potash in mucilaginous liquid, and also employed epispastics. He is of opinion that the alkaline salt diminished the viscidity of the sputa, rendered the cough less harsh and the urine more alkaline. I hope Dr. Popham will pardon me for believing that these supposed advantages are to a great extent imaginary, and that his excellent paper can only be regarded as a valuable contribution, confirming the advantages of the restorative treatment.

An excellent example of a mild mixed treatment is described in a lecture by Dr. Sieveking,[4] who, in opposition to the views I have advanced, and the restorative treatment which has been proved to be so beneficial in pneumonia, lays down for his students two principles. These are, first, that pneumonia is not an entity, and second, that pneumonia differs in type at one and the same time, and therefore demands a varying treatment. As this last idea still extensively prevails among medical practitioners, it may be useful to analyse the evidence furnished by Dr. Sieveking of its correctness. It consists of four cases, very imperfectly recorded.

Case I.—A robust man, æt. 26, admitted on the sixth day with pneumonia of lower half of right lung posteriorly. The treatment was confinement to bed and low diet. On the thirteenth day there was debility, for which quinine and ordinary diet was given. On the seventeenth day he was discharged well.

Now I have little doubt, and the cases I have recorded prove, that if this robust man had been well supported from the first he would have recovered much sooner, and that the quinine was altogether unnecessary.

Case II.—A healthy man, æt. 22, admitted on the seventh day with double pneumonia at the bases, but to what extent is not stated: had marked dyspnœa, and other apparently urgent symptoms. He was bled by venesection to six ounces, and an acetate of ammonia mixture ordered, containing $\frac{1}{12}$ gr. of antim. tart. for a dose, to be taken every three hours. On the following day there was great relief, and the disease was “knocked down,” although it is stated that dulness over the bases continued. He was dismissed cured “a few days” afterwards.

We have here the dyspnœa so commonly present in cases of double pneumonia on the sixth or seventh day, which readily disappears by itself, and is relieved by a warm poultice. It is supposed, however, that a small bleeding of six ounces “knocked down,” or, as some call it, jugulated or strangled the pneumonia. What really happened, however, was that the dyspnœa and apparently urgent symptoms disappeared on the eighth day, which is the usual occurrence. It is distinctly stated that the lungs remained consolidated, so that no impression was made on the disease. What is meant by being dismissed “in a few days” it is of course impossible to tell.

Case III.—A girl æt. 15, admitted on the eighth day, with double pneumonia—the left side more affected than the right, but the extent on neither side stated. She was ordered mist. ammon. acet. with small quantities (?) of morphia. Two days afterwards articular rheumatism appeared. On the following day six leeches were applied to the left side with marked benefit, and a small quantity (?) of antimony was added to the mixture. Dismissed cured on the thirty-second day.

Here was a case of double pneumonia and acute rheumatism running their natural course in a weak subject. Is it to be supposed that six leeches to one side modified the one, or that the "small quantities" of morphia and other treatment influenced the other? Would not the course of both have been shortened by a restorative treatment?

Case IV.—A labourer, æt. 23, admitted on the eighth day, with pneumonia below the fourth rib, anteriorly on the right side. "Six leeches with saline mixture, containing $\frac{1}{12}$ gr. of antim. tart., followed by a blister, appeared (!) to give temporary relief." On the twelfth day typhoid fever declared itself, with bronchitis. Brandy, stimulants, and poultices were then ordered. Dismissed cured on the thirty-fifth day.

Dr. Sieveking says of this case that probably the patient might have done equally well without the leeches and tartar emetic. Of this there can be no doubt. The progress of broncho-pneumonia is always more tedious than that of simple pneumonia, and the recovery was further delayed by the complication of typhoid fever. Can the treatment be defended?

How is it shown in these four cases that the pneumonia in all of them was not precisely the same, that it varied in type, or required a different treatment? That it may be complicated with various diseases, and be associated with strength or weakness of the individual attacked, is no proof of any specific change in the disease itself. In this respect it is in no way different at present from what it has ever been. Then, as to treatment, can it be seriously maintained that the low diet in the first case, that the loss of six ounces of blood in the second, or the six leeches and other treatment in the two others, benefited the pneumonia and hastened its resolution? Of this there is no proof whatever. Unquestionably they tended to an opposite result, as would at once be made apparent if Dr. Sieveking, instead of lecturing on four cases, would tabulate one hundred cases so treated, and let us count what follows. I submit, therefore, that the principles laid down by Dr. Sieveking are in no way supported by his own facts; and, as they are directly opposed to the conclusions derived from more extensive data, they offer no evidence in favour of that mixed treatment which seems so reasonable, and is so popular with many members of the profession.[5]

The question of blood-letting as a point of scientific practice has again been raised by Dr. Richardson,[6] who, appealing to that love of authority so powerful among medical men, asks—"Is it possible that twenty centuries were grossly abused by the infliction of what in the present state of feeling, was, on occasions, akin to crime? I believe not." He then proceeds to discuss ten propositions—or, as he calls them, discoveries made by the ancients; and asks with regard to each of them how far the application of them is sound and judicious practice. His conclusion is, that blood-letting is still useful in some stages of typhus fever; in cases where there is sudden tension of blood, of which sunstroke is an example; in cases of chronic congestion of brain; in cases of acute pain from serous membrane; in some classes of spasmodic pain; in cases of sudden arrest of circulation from concussion; in cases of congestion of the right heart; and, it may be, in extreme cases of hæmorrhage. Above all, he says, "I claim for it a first place in the treatment of simple uræmic coma."

It is impossible to discuss at length, in this paper, all the important practical points referred to by Dr. Richardson. But I shall refer to two great principles in modern as distinguished from ancient medicine, which I think must vitiate the most of his conclusions.

1. When the authority of the ancients is invoked to determine any procedure in medical practice, we must remember that their idea of what constituted disease consisted in the symptoms it manifested. When, therefore, a symptom was diminished or removed, they regarded the means they employed as having diminished or removed the disease. That a blood-letting relieves the pain and dyspnœa in pneumonia is an unquestionable fact. If employed early, it is true the symptoms returned, and the remedy had to be repeated; but if carried out on the fourth up to the eighth day, when, according to the extent of the disease, these symptoms usually subside, and the exudation commences to be absorbed, it appeared to act like a charm. It was then said that the disease was knocked down, or strangulated; and if the patient recovered, however lingering was his convalescence, the value of the remedy appeared to be unquestionable. This idea, it seems, still prevails with some physicians, as we have previously seen that Dr. Sieveking instructed his pupils that he had "knocked down" a double pneumonia by a small bleeding, although the condensation of the lungs—that is, the true disease—still continued.

But modern research has demonstrated that there is no relation whatever between the symptoms and the morbid state of the lung, which it is the object of the well-informed physician to remove. It would be easy to show that there are many cases where all the symptoms of a pneumonia have been present, but where a post-mortem examination has proved that there was no inflammation of the lung; and that a still larger number of instances might be cited where fatal pneumonia has occurred without any of its symptoms having existed during life. Such was the unacquaintance of the past race of practitioners with diagnosis and pathology as now understood, that no confidence whatever can be placed on their impressions as to what disorders were or were not benefited by bleeding.

That in certain cases a full blood-letting modifies or cuts short symptoms, I agree with Dr. Richardson in thinking is just as manifest a truth now, as it was to Galen or Cullen. But I claim for the modern physician a knowledge and a power far beyond that of alleviating symptoms: viz., a true knowledge of the lesion which causes the symptoms, and the power of conducting the disease to a rapid favourable termination, notwithstanding what appears to the inexperienced or uninformed the most alarming and fatal phenomena. He is enabled to watch with accuracy by means of his stethoscope the removal of the consolidation of the lung, to favour the resolution of the exudation, and to assist the excretion of the absorbed products from the economy. These are the aims of the modern practitioner—not so much the alleviation of symptoms as the removal of the morbid state—not soothing his patient, but saving his life. That he is capable of doing this by studying pathology and disregarding the authority of the ancients is no longer a matter of opinion, but is positively demonstrated, by attending to the other principle, which also is not referred to by Dr. Richardson.

2. In all the circumstances which Dr. Richardson thinks blood-letting still useful, we have no solid foundation on which the practitioner can repose with confidence as a general rule of practice. To refer to the opinions of the ancients is, as we have seen, useless; and to support their notions by citing one or two exceptional cases is of no advantage whatever. Indeed the quotation of successful cases, without also stating the failures that have been experienced, is the chief cause of the imperfection of practical medicine. It has been demonstrated that when the practice of bleeding in acute pneumonia was universal, the result was one death in three cases. That was what occurred in the carefully diagnosed and picked cases of M. Louis, as well as what happened in our best hospitals. In those days practitioners triumphantly pointed to the two cases out of three that they snatched from what was then considered a fatal attack of illness. Indeed it

might easily be shown that the worst practice might be defended by what are called successful cases. So far from two survivors out of three being good practice, we have seen that the abandonment of blood-letting and the adoption of a restorative plan of treatment has resulted in diminishing the mortality to one in twenty-seven or thirty cases.

What I object to in medical literature is that prevalent kind of writing, which consists of plausibilities supported by successful cases. What we have at present a right to expect in the way of generalization or theory is that it should be based on positive researches, and not on fallacious authority; and as regards practice, we should have a reasonable number of cases recorded, in which the failures are given as well as the successes. To say that this or that treatment is good, because this or that case recovered, is of no advantage to medicine, unless it stimulate the practitioner to record his cases, tabulate the result, cease from vague opinion, and demonstrate the exact ratio of his success. It is satisfactory for the present state of medicine that such is the kind of inquiry now prosecuted by our most intelligent physicians.

When, therefore, Dr. Richardson is of opinion that a restoration of blood-letting is useful in some stages of typhus fever, and other circumstances previously referred to, I venture to think he should show how the mortality of that disease would be diminished thereby, when contrasted with the nutrient system of management introduced by Dr. Graves of Dublin. The same argument refers to other cases he has referred to. I believe with him that there are instances of uræmic coma, in young and vigorous subjects, which may be cured by blood-letting, but as we have not yet accumulated a sufficient number of such cases it would be premature to speak confidently of the results.

But with regard to the treatment of acute pneumonia I regard the following axioms as fully established, viz.:—

1. The great end of medical practice is to remove the consolidation of the lungs and restore those organs to their natural condition as rapidly as possible.

2. To this end everything that diminishes vital strength should be avoided, and nutrients administered as early as possible, to favour the cell transformation necessary for removing the exudation from the lungs.

3. There is no relation between the violence of the symptoms or force of the pulse and the fatality of the disease. Young and vigorous subjects suffer most, but almost always recover soonest.

4. The weak pulse, want of reaction, non-disappearance of the pneumonic consolidation, or its appearance during the progress of exhaustive diseases, are the unfavourable signs of pneumonia.

5. Continued exercise or work after the attack; low diet, large blood-lettings; depressants, such as tartar emetic and sedatives; expectorants, such as squills and ipecacuanha; mercury and violent purgatives, are opposed to the restorative treatment of the disease, and when not fatal, tend to prolong its duration.

6. Small blood-lettings of from six to eight ounces may be used in extreme cases, more especially of double pneumonia or of broncho-pneumonia, as a palliative to relieve tension of the bloodvessels and congestion of the right heart and lungs.

7. Local pain is best relieved by large warm poultices.

8. The true disease, that is, the exudation which has infiltrated itself through the pulmonary tissues and been coagulated, constituting hepatization, can only be removed, first, by its transformation into pus cells; second, by the molecular degeneration and liquefaction of these; third, by absorption into the blood; and fourth, by excretion of the exuded matter in a chemically altered form through the evacuations.

9. These processes are favoured by supporting the vital powers: first, by rest in bed immediately after the attack; second, by beef-tea and milk during the febrile period, with a moderate amount of wine, if the pulse be feeble; third, by beefsteaks and solid food as soon as they can be taken, with more wine or a little spirits, if the pulse falter; fourth, by mild diuretics on the seventh or eighth day, to favour excretion by the kidneys.

10. The same pathology and principles of treatment apply to all simple cases of pneumonia, whether single or double—the latter being only modified by the weakness of the patient, when more restoratives and stimulants are required.

11. In complicated cases other treatment may be required, according to the circumstances of the case; the pneumonia, however, being always influenced in the manner previously detailed.

12. Since I commenced the treatment of pneumonia by restoratives on the principles just detailed, in 1848, 153 cases of the acute form of the disease have entered my clinical wards in the Royal Infirmary. Of these 129 were simple, and 24 complicated cases. They have been recorded by my clinical clerks, and the progress of each case superintended by my house physician. The whole investigation and the results have been arrived at in public, before successive large clinical classes. Of the 129 simple or uncomplicated cases, of which 35 were double, all recovered, notwithstanding many of them presented the most apparently alarming symptoms. Among the 24 complicated cases were five deaths—1 from ulcerated intestines, 2 from cerebral meningitis, and 2 from uræmia following Bright's disease. Of the whole series, the deaths were 1 in 30⅗ cases. Among the simple cases, single or double, the mortality was *nil*.

NOTES ON THE USE OF POULTICES.

BY GEORGE W. CALLENDER,| *Assistant-Surgeon to St. Bartholomew's Hospital*.

Attention is at present attracted to various applications having for their object the better healing of wounds and sores of different kinds; but I should be sorry if, in the search after new, one at least of the ancient remedies should fall into disfavour. It has been asserted that poultices are often used to conceal defects of treatment, a kind of refuge in ignorance of any more advantageous applications, and that they often do positive harm by inciting profuse, and consequently exhausting, suppuration, and, no doubt, it is true they favour the tendency to suppuration which may exist in particular instances, and that they will increase a suppurating discharge when the latter is already established.

Most remarkable results, however, follow the use of poultices in certain cases—of lupus, for example. A woman attends at my out-patient room with ordinary lupus which, when first seen, had eaten away the middle portion of the upper lip, and had encroached upon the septum of the nose. A bread poultice was applied day and night to the ulcerated surface, and she took iodide of potassium. The sore was soon and completely healed. After six months she returned with the disease worse than ever, but it quickly healed again under treatment, and would, I am sure, remain well if she were ordinarily watchful over it. As the iodide of potassium may have had some influence in this instance, its use was dispensed with in other cases. A woman was taken into Sitwell ward with extensive lupus of the nose. It was with difficulty we persuaded her to submit to such simple treatment as the application of bread poultices continuously to the sore; she craved for physic, which was denied her. Very quickly the sore healed, and she left well. It is needless to record other and similar cases which have been treated in this simple fashion with the same satisfactory results.

Some months ago a man was sent to me from Woolwich with an ulcer on the outer angle of the orbit extending to the conjunctival surfaces of the lids; it was irregularly scabbed over. In my opinion, and in this I was confirmed by several of my colleagues, it was an example of so-called epithelial disease; at all events it had been an open and increasing sore for nearly five years, and before proceeding to remove it I agreed to try the effect of some local caustic. To clean the surface a bread poultice was applied, and it mended so much that this application was continued, when great part of the sore healed rapidly, and the remainder cicatrized after being touched with caustic zinc.

All surgeons are familiar with the good results which follow the application of a poultice to an acutely inflamed surface-part. Quite recently a woman has been under my care with inflammation of the tissues about the internal saphenous vein. She has progressed quite well towards convalescence by keeping the limb at rest, and by having the inflamed vein-track covered with a large poultice of linseed meal; no other treatment has been required. It is a common fault, not so much perhaps in hospital as in private practice, not to give a poultice the chance of curing a local inflammation by limiting its application to the part affected. A poultice to be of any use should widely cover the tissues which surround the seat of inflammation; for example, if the hand is inflamed the poultice should not only completely envelope it, but should extend at least half-way up the fore-arm: and this rule holds good especially when poultices are used for superficial or for subcutaneous diffused inflammation.

A little girl I saw recently in Sitwell ward had a fierce attack of inflammation, after measles, which involved one side of her face and neck. As it threatened to lead to suppuration we made three punctures, carried deeply amongst the tissues, and then applied a succession of large poultices to the entire of the affected side. In twenty-four hours the child, from a condition of great depression, was well enough to leave the hospital—the swelling was much reduced, probably by the draining away of serous fluid, but no suppuration was established. I often direct a bubo to be punctured with a grooved needle, the needle being carried across so as to make a double opening; poultices are then applied, and if the parts are moderately rested, the swelling will usually subside; if the bubo is suppurating the same treatment will suffice to evacuate the pus, and this having discharged the bubo disappears, and no trace even remains of the openings through which the pus has passed out. In cases such as those referred to, some without, some with a surface lesion, the mischief is remedied without any suppurative action being set up by the use of the poultices.

It is desirable, when there is much discharge into a poultice, to dust over the skin about the openings whence the discharge issues some oxide of zinc, or some other drying powder; if this precaution is not taken the matter will irritate and probably enlarge the opening, or will produce vesicles, which break and leave excoriations, or painful papulæ on the adjacent integument. It should be remembered also that great heat is not needed with the poultice; it should be comfortably warm to the patient, and should never be allowed to get, by comparison with its condition when applied, so cold as to lessen the temperature of the part.

Ulcers of many kinds will heal rapidly when treated with poultices; and when I use the word "rapidly," I refer to comparative quickness of healing, as ascertained by measuring the chief diameters of the ulcerated surfaces; their progressive over-closing is thus very accurately checked from week to week. This refers more especially to ordinary ulcers, such as result from injuries. A boy now attends in my out-patient room who under this treatment is healing up a sore on the fore-arm, the remains of a bad crushing of the part. Sometimes this healing is hastened by dusting the ulcer twice daily with powder of oxide of zinc before the poultice is at such times applied. In Sitwell ward a woman is just well of a severe phagedænic sore involving the skin over and below the knee. Mr. Cumberbatch, the dresser of the case, kept the parts at rest by swinging the limb, and applied at first an ordinary linseed poultice, then warm water dressing (another form of poultice), and, to expedite the healing of a few remaining sores, some resin ointment. The cure has occupied twenty-six days, a very rapid progress considering the constitutional nature of the affection: no medicine was needed.

I never could understand, seeing it is desirable to keep the parts immediately after an operation warm and quiet, why those objects should not be attained by the use of poultices; nothing I know of is more efficient to lessen the trouble caused by the starting of a limb after amputation, than the weight of and the resistance offered by a large poultice surrounding the stump. But their employment is in disfavour, first from the fear of their provoking recurrent bleeding, although this reckons for little if due care has been taken to have the wound thoroughly dry before closing it, and unless this care is taken there is little chance of its uniting by the first intention; secondly, by the prevailing notion that such union is prevented by the relaxing influence of this kind of dressing. Wishing to put this to the test of experience, the following cases, amongst others, were placed under treatment.

Having occasion to remove the larger portion of the left upper jaw of a female, about forty years of age, I brought together the incised wound of the face with wire sutures, and directed a bread poultice to be at once applied and renewed at intervals. The entire wound united by the first intention. A boy had his hand and fore-arm crushed by machinery, and it was necessary to perform amputation below the elbow. The flaps of integument were carefully adjusted, and the stump was poulticed. On the ulnar side the tissues united without suppuration; on the radial a portion of skin sloughed in consequence of the hurt it had sustained at the time of the accident, and on this side consequently the repair was less quickly completed. I recently amputated at the thigh, on account of strumous disease of the left knee of a boy, and brought the flap surfaces into apposition. The wound was at once covered with a linseed meal poultice. The next day, the stump being swollen, the wire sutures were cut. Bread poultices, and then warm water dressings, were afterwards employed, and the wound healed without any suppuration having been set up by the action of the local remedies. What pus did form was no more than might have been expected from incomplete primary union of portions of the cut surfaces.

I should like to see a more extended trial given to applications which keep a wound warm and moist continuously from the time of the operation. I think their use would give satisfactory results. No doubt they are most serviceable remedies throughout various forms of ulceration, and especially so in cases of lupus.

THE HYPODERMIC INJECTION OF MORPHIA IN MENTAL DISEASE: A CLINICAL NOTE.[7]

BY C. LOCKHART ROBERTSON, M.D. CANTAB., F.R.C.P.

Medical Superintendent of the Sussex Lunatic Asylum, Hayward's Heath.

In the first number of the *Practitioner*, July 1868, Dr. Anstie has published a Paper on "The Hypodermic Injection of Remedies," in which he truly says, that despite the satisfactory working of the method and of the greatly increased power in handling remedies which it gives us, it is still very much unappreciated. Believing that this remark applies even to the employment of the hypodermic injection of morphia in the treatment of mental disease, I venture on this occasion to lay before the Medico-Psychological Association in the half-hour we devote to Clinical Discussion, a brief outline of three successful cases illustrating the treatment by the hypodermic injection of morphia in recent mania, chronic mania, and melancholia respectively.

In October 1861 Dr. W. C. Mackintosh published a Paper in the *Journal of Mental Science* on "The Subcutaneous Injection of Morphia in Insanity," which first drew my attention to this method of treatment. In the Reports of the Somerset Asylum, Dr. Boyd has also recorded his opinion of the value of this treatment in cases of maniacal excitement with sleeplessness, and in that form of destructive mania accompanied with dirty habits.[8]

The detail of the hypodermic method of treatment is carefully stated in Dr. Anstie's Paper, and to this I must refer those who desire farther information regarding it. I use a solution of 6 gr. of the acetate of morphia to the drachm; Dr. Anstie's strength is 5 gr. I always commence with ♏v of the solution (½ gr.), and in only one case out of many hundred hypodermic injections of morphia has any injurious effects followed the remedy thus used.

CASE I. *Recent Mania.*—J. H. W., No. 1,563, female, aged 20, single; domestic servant. Form of disease, acute asthenic mania.

History.—Never had any previous attack. No history of insanity in her family. Has been engaged for some years as a domestic servant. No reason can be given for her illness. It is stated that for the last three or four months she has been strange, and at times depressed, and that about three weeks ago she suddenly became maniacal, and has remained in a state of violent excitement ever since.

Progress.—On admission at Hayward's Heath, on the 22d of March last, she was in a state of the most intense maniacal excitement, and very incoherent. Physically, she was suffering from marked typhoidal symptoms, her pulse was feeble and very rapid, her skin dry and harsh, her lips and teeth covered with sordes, her tongue coated with a thick creamy fur. She refused all food, and had had no sleep for several nights.

Although she could not be prevailed on to take any solid food, she was coaxed at times during the first two days after her admission to take ½ gr. of morphia in a little brandy, but she was almost invariably sick after it; moreover, the excitement continued, and she could obtain no sleep.

On the third day the hypodermic injection of ½ gr. of morphia was commenced, and continued every four hours except during the middle of the night. On the fifth day she was calm, although incoherent, and had slept during the whole of the previous night, took her food well, and had lost nearly all the typhoidal symptoms. Moreover, the irritability of the stomach was completely allayed.

She has since then recovered without a bad symptom, and she is now convalescent.

This case showed in a very marked manner the advantage of the hypodermic injection of morphia over its administration by the mouth in cases, which so frequently occur, of acute mania with marked asthenia and irritability of the stomach, causing refusal of food.

CASE II. *Chronic Mania.*—W. H., No. 950, aged 68, single, groom. Form of disease, chronic mania, characterised by frequent recurrent attacks of maniacal excitement.

History.—Strong hereditary taint of insanity. Nearly all his brothers and sisters are more or less insane or eccentric. Much given to habits of intemperance, but, although often strange and eccentric, was never sufficiently insane to warrant his being placed in a lunatic asylum until he was 64 years of age, when he was attacked with acute mania and removed to Hayward's Heath.

Progress.—During the attack of mania under which he was suffering when admitted into the asylum he was treated with small doses (♏x) of tincture of digitalis every four hours. The symptoms lasted for nearly three months. He was then calm for many weeks. On the next outbreak of mania, equal parts of liq. opii were added to the digitalis, and with a beneficial effect, the attack not lasting so long.

He was thus treated for some two or three years. He generally suffered from three or four attacks in each year.

In April 1868 he had an unusually severe attack of excitement, combined with much noise and destruction of clothing. The usual medicines having no effect, he was treated with the subcutaneous injection of morphia (½ gr.) three or four times in the twenty-four hours, and with marked benefit.

On the recurrence of the next attack subcutaneous injection was had recourse to at once, and the period of excitement was reduced to little over a fortnight.

The next attack passed off in an equally satisfactory manner. In the January of the present year an attack of recurrent mania being evidently imminent, the old treatment of digitalis and opium was tried for fully a fortnight, but without benefit. On February 8th ½ gr. of morphia was injected, and the injection continued every six hours, and on February 10th (to quote from the case-book) he was decidedly improved and free from excitement and noise.

Not only, therefore, is the duration of the attack of recurrent mania diminished in this case, but during the attack the excitement is much less intense under the hypodermic method of treatment.

CASE III. *Melancholia.*—M. T., No. 1,397, female, aged 57, married, domestic servant. Form of disease, acute recurrent melancholia.

History.—No hereditary taint of insanity. Has been insane and confined in asylums three or four times. She is temperate in her habits, and her attacks of insanity appear to have followed on most occasions the puerperal condition, but the present illness is stated to be due to family troubles.

Progress.—On admission she was suffering from the most acute type of melancholia, combined with insomnia, refusal of food, and a strong suicidal tendency. Moreover she was in a poor physical condition, having lost much in weight, and being thin and anæmic.

In the first place she was treated with stimulants, sedatives, and a nourishing diet, but she remained from May 14th, the day of admission, until May 20th, without any improvement, and was becoming so reduced, from want of sleep and constant worry, that her life was despaired of. On the 20th May, 1868 (to quote from the case-book), “she passed a very restless night, and is much exhausted this morning: injected acetate of morphia gr. j, and she soon fell asleep; took her food well on awaking.”

On the 23d, “injected gr. j of morphia twice daily since the last entry, and with decided benefit, and she is much less excited. Sleeps well, and the suicidal tendency seems to have passed away.”

On July 15th the entry is as follows:—“Has improved uninterruptedly ever since the last entry, and is now tolerably sane.”

She was discharged recovered on 7th December, 1868, and has continued sane to this date, although in such a case another relapse is most probable.

ON THE THERAPEUTICAL VALUE OF THE INHALATION OF OXYGEN GAS.

BY EDWARD MACKEY, M.B. LOND. ETC.

Joint Professor of Materia Medica and Therapeutics in Queen’s College, Birmingham.

Our ordinary medicinal agents are substances from the animal, the vegetable, and the mineral kingdoms: the one here to be treated of is of that class of remedies which includes the great elements or forces of nature: such are water, in all the varied forms of bath; electricity, in its different developments; air, in all its modifications of pressure or composition.

It is unfortunate that the application of these mighty remedies seems liable to degenerate into charlatanism: partly perhaps because they have the power—*wrongfully* claimed for quack medicines—of doing good in many apparently different forms of disease; partly because their use must at present be limited to the few, and does not admit of ready introduction into the practice of the many.

Nevertheless, the truthful study of these agents offers scope for the highest science, and promises therapeutical results of the highest value. The following cases are offered as data for judging of the value of one of them. I do not propose to treat here of the chemistry of oxygen,[9] nor of its physiological effects, nor even of the objections which have been urged against its use—but simply to state facts which have come under my own observation.

CASE I. *Emphysema pulmonum (hereditary).*—A lady of 55 had been for many years the subject of constant dyspnœa, increased on all movement, and often amounting to a sense of suffocation. A physical examination revealed sibilant râles with prolonged expiration heard all over the chest, which was of large capacity and more than normally resonant on percussion; the heart's action was weak and the circulation embarrassed, as evidenced by œdema of the face and extremities.

She was subject to attacks of bronchitis occasionally, but, at the time of treatment, the general health was in fair condition; the prominent complaint was the difficulty of breathing.

On July 5, 1868, she inhaled a mixture of 3 pints of oxygen with 30 of air: the results were favourable. Within a few days the dose was doubled, 6 pints to 60: soon the proportion of 8 to 60 was used: and later, 12 to 60, and with this dose we seemed to obtain such good effects that I did not think it necessary to increase it. The inhalations were taken at intervals of three or four days for a space of six weeks; after each one, the lady experienced marked relief, which she expressed as being able to take a deep breath and get sufficient air—a feeling not known for years; as being able to move with comparative ease, feeling buoyant, and more like healthy persons should feel, than she ever remembered.

The only definite effect upon secretion was a more copious and facile expectoration, always produced, and lasting for a day or two; the effect upon the circulation was not marked at the time, but some palpitation occurred, generally in the nights which followed the taking of the larger doses; no other unpleasant symptoms whatever.

In attacks of exaggerated dyspnœa, as they occur sometimes in the emphysematous, and in those peculiar, nervous, irritable states apt to be induced by mental causes in the subjects of weak hearts, I have known her come into my consulting-room, inhale for half an hour, and express herself cheerful and composed. Nor was this the effect of fancy; for, at first, the lady had a prejudice against the plan; now she esteems it highly, nor has she ever found relief at all comparable to this, from the many medicines prescribed at various times by various practitioners.

Case II. is of the same nature, and occurred in a gentleman of 24, who had had good health till twelve months before, when he noticed for the first time wheezing, and afterwards cough, traceable partly to the dusty nature of his business, partly to wearing damp clothes. The chest symptoms continued so bad as to confine him to the house for three or four months; afterwards, he seemed gradually to recover under the use of tonics and cod oil, and the influence of a warmer climate, and when he came to me in January 1869 for the first time, he looked well; however, he complained of debility, of constant dyspnœa on exertion, and of exaggerated attacks of it occurring suddenly at times, of some cough and of glutinous expectoration; if he attempted to live well, as he had been told to do—meaning especially the taking of wine—he usually got an attack of epistaxis.

Physical examination revealed a sibilus at the end of inspiration, and a rhonchus with expiration over all the right lung, except the apex; the chest was very fully developed and abnormally resonant.

I prescribed for him inhalations of oxygen in the proportion of 12 pints to 60 of air, and he took these twice in the week for five weeks; after each one he expressed himself in much the same manner as the last patient, was conscious of a general feeling of renewed health, of a greater power of breathing, and of facility of expectoration; great improvement took place in his condition, and I think it must be credited principally to the gas; for, although I ordered him 10 to 20 drops of tinct. lobeliæ at night-time, and later on tinct. fer. acetatis and frictions with the linim. tereb. acet., yet it is to be borne in mind that he had previously had a fair trial of expectorants, tonics, and even change of air, without anything like equivalent relief.

Case III. *Phthisis pulmonalis.*—Mrs. W——, æt. 31, who had lost her father and sisters of consumption, consulted me in Dec. 1867. For the last six months had had cough, for the last three had emaciated, and at this time had the prostration, night sweats, diarrhœa, and hectic of the third stage of phthisis; hæmoptysis had occurred several times: the expectoration was generally purulent. There were violent pains, especially over left chest, and examination revealed a fine crepitus at apex of left lung. The patient was treated with ordinary medicines, and improved gradually. Opium in the form of an atomized spray was found to be the best medicine for relieving cough, and procuring sleep; tincture of steel and carbolic acid used in the same manner relieved, to a certain extent, the profuse expectoration; and although the case became complicated with a peri-uterine hæmatocele, in February 1868 she rallied from this also.

It was July 1868 before she could walk as far as my house. Her principal symptoms then were debility, pains in the chest, cough, and copious muco-purulent sputum. At this time she began inhalations of oxygen in the proportion of 6 pints to 60 of air, increasing by degrees to 12 pints. She took eight inhalations at intervals of two days, and then found the above symptoms so much relieved as to be able to omit all treatment for a time.

She herself attributed great benefit to the gas, and was taking no other special medicine at the time. Since then she has borne fairly well the cares of a large family. She has gained flesh, and though there is still a frequent cough, and sputum, and a mucous râle about the left apex (I examined the chest two days ago), the progress of the disease is arrested for a time at least.

Case IV. points precisely in the same direction. In May 1868 I was consulted by a gentleman of 19, whose father died of phthisis. He had been steward on board a packet plying between Liverpool and New York; got wet through on his last voyage, lung symptoms soon set in, and he considers that his present ones date from three months ago. He has constant cough, for which he can get no relief, profuse sweatings, hectic, and extreme emaciation; in short, all the ordinary signs of softening tubercle in the right apex, and had been sent home by medical men in Liverpool to Handsworth, —just to die. However, he too rallied under careful nursing, and with the help of ordinary medicinal agents, and by July was able to walk to my house, and begin inhalations of oxygen in proportion of 6 pints to 60. At this time the above-named symptoms were all better, and his principal complaint was of difficulty of breathing, and of pain in the side of the chest, and these did not yield to medicines or to liniments. He continued to inhale twice a week for two months, and at the end of that time was sufficiently recovered to seek for a situation. He is now in the employ of the London and North-Western Railway Company, has gained two stone, he says, and is 6 ft. 4 in. in height. I had an opportunity of examining his chest last week, and detected only dry and interrupted respiration in one apex. I should add that he continued the tinct. fer. perchlor. and cod oil during and after his treatment by gas; but he distinguished relief to the dyspnœa from the gas alone.

Case V.—Rev. W. M——, aged 34, lost father, brothers, and sisters from phthisis. In February 1868, when I first saw him, the prominent symptoms had lasted six months—the dyspepsia, the tight cough, the loss of voice, and the emaciation.

In March the physical signs of phthisis were evident in the left apex, as was ascertained by Dr. Russell, who saw the patient with me at that time. I need not detail symptoms or treatment, as they did not differ from what is usual; suffice it to say that improvement took place, but was temporary, and in April we recommended him to visit Jersey. He was there for three months (being considerably longer than I had intended), and at that time he thought that he found benefit from the sulphurous acid spray. However, he returned as bad, if not worse, than when he went, with night sweats, extreme prostration, cough, difficulty of breathing, and purulent expectoration. It was in this condition, and when he had had a trial of almost every other remedy, including a prolonged course of cod oil, that I proposed oxygen to him, and he began it July 24, 1868, in proportion of 6 pints to 60, increasing gradually up to 10 to 60, and taking this two or three times a week up to October 8, a period of 2½ months; during the whole course of the time, he had expressed himself as much relieved, both as to breathing power, cough, character of expectoration, appetite, and strength. He had gained weight, and the malady was quiescent. He had been accustomed to come from the country by train, and to ride back in a cab. On one unfortunate day (October 8), which was cold and very wet, he got into a cab the window of which was broken, drove six miles in the night air, in the course of that night got a sudden pain in the side, and dyspnœa, and when I saw him next day pneumonia had attacked the right lung, and he was desperately ill.

Now the point of the case is this. It has been said that the inhalation of oxygen is liable to cause inflammation of the lung. Did it do so in this patient? That must be a question to be decided on the evidence, but I cannot think that it did. The dilution of the gas was great; the same quantity had been inhaled for weeks before without any injury, and the other exciting cause was such a probable one. At the end of a month's time he was convalescent, and urgently requested the resumption of his inhalations. I consented, and he again expressed relief from them, especially as to the dyspnœa; but effusion in the right pleura came on gradually, but too surely; for some time we saw the end approaching, and he died last month. Almost to the last he expressed benefit from the gas, and he certainly suffered less than any consumptive patient whom I have ever seen.[10]

Case VI.—I adduce as an instance of another variety of dyspnœa a warehouse woman of 27, who had also lost several brothers and sisters of phthisis. Had been much depressed by nursing the last one through a long and painful illness; she came to me in June 1868, with symptoms of dyspepsia and history of attacks of urgent difficulty of breathing coming on generally at a fixed hour of 9 or 10 in the morning; occasionally after later meals; she kept constantly sighing deeply, and had various symptoms of hysterical temperament; had also cough and viscid expectoration; but a physical examination revealed nothing very definite—perhaps puerile respiration in one lung and diminished vesicular murmur in the other.

She was treated for some weeks with various stomachic and tonic medicines, and went into the country for a fortnight; but the symptoms remained more or less.

It was during an attack of this spasmodic or hysterical dyspnœa that I first administered oxygen to her, in proportion of 5 pints to 30 of air, and again in a double dose, only on three or four occasions.

It is possible that these doses were not large enough for a fair trial; but, however, I wish to record that relief was given, but it was slight and not permanent. Eventually the patient recovered under the use of bromide of potassium and quinine. She has since married, and is well.

Case VII. *Chlorosis.*—Miss P——, æt. 21, had been employed for some years with very long hours of work in a small close room; was stunted in growth, with chlorotic complexion, drowsiness, headache, palpitation, dyspnœa, and great fulness of the thyroid gland. Menstruation still occurred, though scantily, and at intervals of six to ten weeks.

She came under my care in January 1868, and after regulating her hours and her food as far as possible I prescribed various preparations of iron, of magnesia, of aloes and myrrh, baths, and liniments, &c.

There was a gradual improvement in her condition with occasional relapses, which obliged her to remain under treatment for many months. In October 1868 the prominent symptom was headache, violent and pulsating, and it was for this that I prescribed oxygen, after the failure of many medicines.

November 8th.—Inhaled 6 pints diluted with 30 of air; pulse 96 before commencing, had same frequency at the end; the only special symptom felt was a sense of oppression at the chest, but the headache was not so bad as usual that night.

9th.—When she entered my room the headache was violent; she inhaled 12 pints in 60 of air, and before finishing, the headache had disappeared, and she felt better; this occurred on several *though not on all* occasions, but she continued the use of the gas for ten days only. She was then obliged to be away for a time, and the headaches returned shortly afterwards: relief had been given, but not permanently. Perseverance here *might* have shown good results, but an opportunity occurring soon afterwards of a residence in the country for some months, I recommended her to take advantage of it; she has recently returned much improved in health, and is following her occupation again.

The two following cases are somewhat allied to the last, in being disorders of secretion or excretion, and are examples of that so common complaint in town people, hepatic congestion; the results were exceptionally favourable, and although under ordinary medicines patients generally improve in a satisfactory manner, yet the malady sometimes shows such a disposition to return, that one really scruples about prescribing over and over again rhubarb, magnesia, nux vomica, acids, or blue pill.

Case VIII.—Mrs. B——, æt. 40, mother of a large family, had been subject to bilious attacks for many years, and had several times been under my care. In May 1868 she had pain over hepatic region, depression, nausea, headache, and yellow conjunctivæ; the stools were frequent, loose, and pale; menstruation was profuse, and occurred oftener than normal.

She took at first dilute acids with tinct. rhei co. and improved, but relapsed at the end of June, and it was then that I recommended the gas to her; the prominent symptoms being headache, depression, complete loss of appetite, and a constriction about the chest "as if she could not get air enough:" menorrhagia had been going on for two days.

June 23d.—Inhalation of 4 pints in 60: there was not any marked effect.

25th.—6 pints in 60, and before the inhalation was over the headache was relieved, and all that day she felt "lighter" and better, though rather strange; to bed early, and slept and woke without headache, the first time for nearly twelve months, and was nearly free from her shoulder pain.

27th.—Dose repeated with similar good results, and no medicine taken; the diet was regulated as it had been before. To be brief, she took eight inhalations on alternate days, and at the end of that time was well enough to do without treatment: not that she was quite well, but restored to her ordinary health, and the improvement has up to this time continued.

Case IX.—A lady of 21, after a period of great mental anxiety and of close application to business, began to feel extreme depression, drowsiness, anorexia, headache, nausea, and interscapular pain; the pulse was slow, the face pale; there were palpitation and dyspnœa without signs of organic disease. The symptoms had lasted about two months, when I first saw her in July 1868. She took alkalies, aperients, and appropriate medicines, and on the 12th took inhalation, 7 pints in 60. Here again the same remarkable effect was produced, in relieving headache before the end of the quantity. She continued to inhale a little larger dose every third day for a fortnight, without taking any medicine for the latter part of the time. She has remained fairly well ever since, and voluntarily expressed the great benefit which she derived from the gas, especially as to relieving a sense of constriction across the chest and dyspnœa.

Both these cases were tolerably acute, and occurred in persons of naturally "sanguine" temperament, but it is necessary to record that another case which I have treated more recently,—a young lady of "bilious" temperament, who suffered from hepatic congestion in a more chronic form,—found no special effect whatever from inhalations taken on alternate days, for a fortnight.

Case X. was one of albuminuria in a lady of 57. The disease had commenced four years before, and her health had been markedly impaired for the last twelve months (dating from an attack of vertigo, and loss of consciousness). Last winter she had had bronchitis. She was a lady highly connected, and had been under the care of several eminent London physicians, who had concurred in advising her to go into the country for a time, and I was sent for to see her in November last, when she had already been at a country-house in this neighbourhood for some months.

She was feeble, with pallid face and injected cheeks; extremities œdematous; dyspnœa to a great extent on the slightest exertion; tendency to fainting and giddiness; urine deposited urates, and gave a cloud of albumen with the usual tests. Almost the only remedy which had not been given to her was this gas. I requested her to write and ask her physician if he concurred in its use; he wrote back to say "by all means," and on November 18 she began with 14 pints in 60 of air. The pulse was 78 at commencing, and did not vary. She took it six times at intervals of three days. I had anticipated good from it, but there was really no marked effect. She thought, in fact, that her headache was rather worse afterwards, but I think that was better accounted for by the carriage drive to my house and the extra excitement.

Treatment was omitted for a time, and in the interval she got an attack of subacute bronchitis; on recovery she hired an apparatus of her own, and began, on December 12, 16 pints to 60. I consider that she had a week's fair trial, but at the end of that time, what with leakage in the machine and non-arrival of gas, the lady's patience failed, and the treatment was not persevered with. I mention these matters as an instance of one of the difficulties that an unusual mode of treatment must necessarily contend with. However, the result of this treatment, such as it was, gave no encouragement to persevere.

Case XI. resembled the last in the fact of there being organic disease. She was a delicate and refined lady, of the age of 34, unmarried. With a history of some years of spinal debility, and of congestive headache, at this time (September 1868) there was general prostration, numbness, and tingling in various parts, a sense of suffocation and of constriction, and partial loss of power over limbs; but worse than all, the attacks of headache of frightful intensity, attended with throbbing, flushing, and confusion of thought, and generally located over the left eye, which then protruded very much. Of these and other symptoms, some were explicable on the hypothesis of congestion of the spinal cord, and parts of the cerebrum, while some suggested a grave suspicion of ramollissement; others, again, of a chronic thickening of the membranes. My opinion was necessarily doubtful, but with regard to remedies oxygen offered a prospect of relieving at least some of these symptoms; it is said to have done so in recorded cases. Moreover, the patient had had the best obtainable advice in her own town of Wolverhampton and in the city of Cork, remedies prescribed by her physician had not benefited her, and for some months she had been under homoeopathic treatment at home and at Malvern. I recommended her to hire an apparatus for her own use; and on October 4 she began with 12 pints to 50 of air, the inhalation to be extended over the period of one hour. The necessary exertion tired her, and she felt no appreciable relief. I did not like to wait longer without attempting to relieve by some of our usual remedies, and I prescribed gr. viij pot. brom. with ♏ viij. liq. ergotæ, as having a special influence in equalizing the spinal circulation.

Oct. 12.—Head bad, but not so bad as usual. On 15th, menstruation came on, and aggravated symptoms somewhat.

I directed inhalations to be increased in strength every day, until I reached equal proportions of air and oxygen—as much as 30 pints of each.

On October 21 had an attack of prostration to a more extreme degree than ever known before; she seemed, in fact, at the point of death from sheer exhaustion. And here again we are met by the important question—Was this due to oxygen? for experience recorded of its effects seems to warrant this apparent paradox, that although in many cases a stimulant, in some it is a depressant; that although it will increase the vital powers when only moderately depressed, it will tend to lower them when they are already very much lowered.[11] Or, again, was the prostration due to the bromide of potassium?

Candidly, I do not think that it was due to these causes, partly because she had had no inhalation for two days before, and no medicine for three days, and partly because a depression similar, though less in degree, has followed menstruation on other occasions, and this had been more profuse than usual.

For the time I gave her quinine and brandy and a little morphia, and on the 23d permitted her to resume inhalation, beginning at 12 to 60; she again gradually increased the dose to 30 pints in the day. For the bromide of potassium I substituted small doses of strychnia. The administration of the gas in varying doses was persevered with till November 4—a full month altogether—then I recommended her to discontinue it. The effect was certainly not marked; if there was any, it might be in relieving the sense of suffocation, which was not so bad during that month as it had been before and since; but on the whole the gas must be considered to have failed in this case. However, it will be remembered that many other remedies had failed also, and the further progress of the case has convinced me of the presence of serious organic disease; it is in fact two months since I have ceased to entertain or give any hope whatever of this lady's recovery.

Case XII. was one of general debility with irritable heart. A gentleman of 35, who had lost several brothers by phthisis, and had been subject to unusual harass and exertion, began to lose appetite, to grow thin, and to suffer from lassitude, dyspnœa, and palpitation. When he came to me in May 1868, the symptoms had lasted for two or three months, but I could detect no physical signs of disease. For some weeks he took quinine, aconite, and cod oil, and applied belladonna; still he did not improve much.

On June 21, he inhaled 4 pints mixed with 30 of air, and felt a “greater lightness”—no increase of palpitation. After four days of treatment, he got an opportunity of spending a fortnight in the Highlands, and I recommended him to try the breathing of oxygen there. He returned home, however, in July, not so much improved as we had hoped, and still complaining much of soreness about the chest, and oppressed breathing. From this time to September he took an inhalation every third or fourth day, and with perceptible benefit. It is true that he took, for some weeks of the time, the hypophosphate of lime and cod oil, but still the effect of the inhalation in improving breathing power, and appetite especially, was immediate enough to convince us that it had a large share in his recovery. He has remained fairly well since.

To resume: 12 cases are here related; 2 of the 12 are of organic, and in all probability incurable disease, and these 2 derived little or no benefit from the inhalation of oxygen; the other 10 found benefit as recorded, some more, some less, but all of a kind which I have not seen given by medicine alone. It remains to ask—Is there any common character by which we may connect together this series of cases, and which may enable us to say, oxygen is good for such and such a class of cases, as we say iodide of potassium or quinine is good for such and such a class?

I think that we may find some such common character in the presence of congestion, especially venous congestion, whether of the liver, the lungs, or the uterus: more than this I will not say at present; the classification of carefully-observed cases, and a rational theory of this “modus medendi,” are points that require special study, and cannot be dogmatised upon until we have a wider basis of facts.

ON THE HERPETIC FORM OF STRUMOUS OPHTHALMIA, AND ITS TREATMENT BY ARSENIC.

BY ROBERT S. OGLESBY,
Assistant Demonstrator of Anatomy in the Leeds School of Medicine.

Strumous Ophthalmia, associated with herpes of the face, or as it may be termed the herpetic form of strumous ophthalmia, is a disease so often met with in general practice, and one so little amenable to local treatment, that I venture to say a few words regarding its treatment constitutionally. I should hesitate to occupy valuable time and space with what appears to be a trivial subject, on which much has already been written, did I not believe that by so doing additional light might be thrown upon the subject. For several years past I have been collecting evidence, carefully sifting and placing all well-marked cases aside for special treatment. The results obtained in the earlier batches of cases thus treated, decided me to continue such treatment for a lengthened period. The evidence thus obtained being highly satisfactory, other treatment formerly employed was abandoned.

During the time that has since elapsed, I have continued to employ the same remedy with the same good results.

This form of the disease generally presents itself in fat, red-faced children who have the appearance of robust health. On questioning the parents, they will perhaps inform you that the child was but small and puny at birth, and for the first few months of its life it was sickly and delicate, and not until lately had it become so stout and healthy. They attribute the change to the purer air the child now breathes, for they have removed from a populous and unhealthy district to one less crowded and decidedly more healthy.

In such a child we find traces of constitutional defect in enlarged and rickety joints, a head big and ill-shapen, and an anterior fontanelle not completely closed. We find a thick and prominent lip, conspicuous for the extent of mucous membrane it shows.

The intolerance of light is so intense that the child cannot bear even a moderate degree, but persists in burying its face in its hands, or hiding from the light in some dark corner. But what strikes us so forcibly, and what really is so characteristic of the case, is the fact that the patient's face is disfigured with patches of herpes,—a fact which makes us hopeful, as these cases are as a rule the most amenable to treatment. To cure the disease of the skin is to cure the disease of the eye in the most rapid and satisfactory manner.

In the majority of the cases which have come under my care, the eruption was confined to one-half of the face below the brow. In a small proportion the side of the nose was not affected, and in several the skin of the upper lip and chin escaped altogether. The vesicles in most of the cases appeared to follow the course taken by those branches of the infra-orbital nerve which supply the skin of the face.

My notes do not supply me with any case where the eruption invaded the brow, although I may have overlooked some such case in my earlier investigations on the subject. The eruption was often accompanied by a febrile condition more intense than I have ever met with in the other forms of strumous ophthalmia.

Before proceeding to active treatment, instructions regarding diet, regulation of the bowels, &c. should be given. The diet should be plentiful, simple, and nutritious; and all articles of food likely to unduly tax the digestive powers (which are as a rule weak in such children) should be carefully avoided. Strict attention should be paid to the bowels, which ought to be opened at least once during each day, but oftener if the appetite be faulty, the tongue loaded, and the fæces light coloured and of bad odour. These preliminary instructions having been attended to, special treatment may be adopted.

It is well to begin with small doses of the arsenic in form of Fowler's solution. Two drops may be given thrice daily, in some bitter infusion, to a child between one and two years of age, and gradually increased to four drops. Seldom is it requisite to further increase the dose. Arsenic appears to exercise a marked control over the febrile symptoms of this disease. As the herpetic eruption diminishes, the child ceases to shun the light, and as the rash fades the pustule on the eye heals. The benefit of fresh air in the more obstinate forms of the disease is well known; but it is often difficult to convince parents that exercise in the fresh air will benefit the child. They imagine that it is rather hurtful than otherwise, because the intolerance of light is then a distressing symptom, proper means not having been taken to shade the eyes. A ready method is to place over each eye a pad of cotton wool, and over the pads a bandage, which should encircle the head, and tie on the occiput. The pads should be frequently renewed and the eyelids washed with warm water.

In a future paper I hope to bring forward additional facts on the subject, and conclude by giving the history of a number of cases.

Reviews.

Klinische Beiträge zur Erkentniss und Behandlung schwerer Krankheitsfälle. Von Dr. Adolf Hermann. Primararzt in Pest. Wien: W. Baumler, 1868. Gr. 8vo. pp. 282.

(*Clinical Contributions to the Diagnosis and Treatment of severe Cases of Disease.* By Dr. A. Hermann.)

The preface to this work explains that the author enjoys, at the “Israelitenspital” of Pesth, a considerable field of experience, but is less overburdened with cases than are many physicians of hospitals where people of all creeds are received, and thus has time to study them individually with the greater care. We gather that this volume, which is the fruit of official labours during two years, is the first of a series which it is intended to publish, and it fairly enough fulfils the promise which the preface holds out. One of the most interesting parts of the book is the observations on “tuberculous” affections of the larynx. We must say, however, that the general treatment of the subject of tuberculosis disappoints us, and certainly does not come up to the standard of accurate and careful work which the author has set up for himself. It shows few traces of that wide acquaintance with modern pathological researches on its subject which is more than ever essential to the clinical observer who would throw any light, even by means of the most diligent labours, upon those matters which are really the *quæstiones vexatæ* in regard to the nature and treatment of the various diseases commonly confounded under the name of tubercle. The last chapter in the book contains an interesting account of the author’s experience of the hypodermic injection of remedies. We observe with surprise that he comes to conclusions very unfavourable to the subcutaneous use of atropine, which he almost totally condemns. We entirely agree with him in the statement that doses as high as half a grain, or even less, will at times produce some cerebral and vaso-motor symptoms, but the persistent use of much smaller quantities does, we believe, meet with quite a different and a much higher measure of success. On the other hand, the author speaks with the warmest praise of the efficacy of hypodermic injection of morphia in all kinds of neuralgias; for these diseases he declares that there is no remedy comparable to it.

Annuaire de Thérapeutique, de Matière Médicale, de Pharmacie et de Toxicologie, pour 1869. Par A. Bouchardat.

M. Bouchardat's well-known compact little yearly volumes are always welcome, and always useful; and this year the *résumé* includes a rather unusual number of interesting matters. The first thing which deserves notice is the recent researches on the therapeutic action of arsenic in phthisis; a subject which we have for some time past desired to discuss in this journal, but have been prevented by want of space and time. The very able paper of M. Moutard-Martin, read before the Academy in November last, called forth a report from M. Hérard, which speaks in such strong eulogy of the remedy as used in the manner and under the conditions laid down by M. Moutard-Martin, that we are considerably surprised to observe the small amount of notice which has been given to the subject in England. Arsenic has of course been long known as a tonic more or less applicable to phthisis, as to other states of debility. But the points so sharply brought out by the French author—the limitation of the therapeutic action of arsenic to the non-febrile periods and cases, and (on the other hand) its extraordinary efficacy within these limits, in restoring flesh and strength and general health, and wonderfully amending the state of the lungs themselves—are so important that they demand the serious and immediate attention of English physicians. M. Moutard-Martin employs the remedy in pills, as being more convenient than the liquid form; each of these pills (*granules de dioscoride*) contains a milligramme (·00156 of a grain) of arsenious acid, with manna and honey.

Another matter which deserves notice, and had escaped ours, is the experience of M. de Beaufort as to the efficacy of iodide of potassium in diseases of the lachrymal apparatus. This observer began by applying the treatment to comparatively recent and slight cases of obstruction of the sac and nasal duct, the result of coryza, chronic conjunctivitis, &c. Obtaining a speedy cure in several such cases, he proceeded to try it even in instances where a tough fibrous stricture had existed for a long time. Even in such apparently unlikely circumstances, he has twice obtained success by the treatment. Where there is lachrymal fistula, following an abscess of the sac, the iodide is also very useful, but its employment should be accompanied by applications of tincture of iodine.

A matter of some consequence is the opinion of M. Regnault respecting the comparative activity of various preparations of digitaline. In his preface to the seventh edition of Soubeiran's *Traité de Pharmacie*, this author remarks on the serious difference which exists between various samples of so-called digitaline. He declares that he finds himself driven to employ exclusively the *granules* of Hornolle and Quevenne. Under the head of digitalis also we may notice the employment, by MM. Dumesnil and Lailler, of a combination of opium (in the form of *extrait gommeux*) with tincture of digitalis. This appears, according to the authors, to have a calming effect upon the excitement and sleeplessness of insane patients, with less tendency, at the same time, to congestive effects than is the case when opium is given alone. They say that not merely does it produce better immediate effects, but that it can be more harmlessly continued after the original excitement has calmed down than other narcotics. But they remark that as digitalis, in therapeutic doses at least, has a well-known tendency to wear out the susceptibility of the organism for it, it is best to suspend its use as soon as may be. Two formulas are employed by the author:—

POTION NO. 1.

Extrait gommeux d'opium	⅓ grain.
Tincture of digitalis	7½ minims.
Syrup	1 ounce.
Distilled water	5 ounces.

POTION NO. 2.

Extrait gommeux d'opium	⅔ grain.
Tincture of digitalis	15 minims.
Syrup	1 ounce.
Distilled water	5 ounces.

According to circumstances, the stronger or the weaker of these two prescriptions is the allotted potion for twenty-four hours. We have recounted all this gravely, but must not be expected to preserve our gravity to the last. Not even the respected name of M. Dumesnil can keep down an irresistible tendency to laugh when we are seriously told that 7½ minims of tincture of digitalis, taken in two separate doses in the course of twenty-four hours, will avert the mischievous excitement which might otherwise be caused by ⅓ grain of extract of opium similarly distributed! That neither potion No. 1 nor potion No. 2 can be considered a dangerous narcotic we quite allow; but we fancy that must be because there is so little opium in either of them, not because there is so much digitalis. In fact, if we might be allowed to make a delicate suggestion to our therapeutical brethren of *Outre-mer*, it is, that just now they are going the least bit in the world crazy over digitalis and its wonderful properties. However, we have nothing but praise for such researches as those of MM. Oulmont and Hirtz, already noticed in these pages.

A small matter worth noting is the suggestion of M. Hager as to the bad effects of *impure glycerine* which are occasionally met with. It appears that certain glycerines, which are locally irritant when applied to the skin, prove to contain formic and oxalic acid; the latter is more especially the irritating agent.

Some interest attaches to a comparison instituted by M. Rabuteau between the action of sulphate of *soda* and that of sulphate of *lithia*. The former diminishes or removes thirst and produces constipation; the latter increases thirst considerably, and causes copious liquid stools and watery vomiting. The soda salt *dries* the alimentary canal, the other thickens the blood and pours out its watery element in abundant intestinal secretions.

The chlorides of sodium and lithium present analogous differences. The former does not purge when introduced into the veins, though it does purge when given internally in similarly large doses. Its ormolic effects are like those of sulphate of soda. Iodide of sodium has similar differences of effect, according as it is given by the veins or the alimentary canal. In short, the purgative effects of salines would appear to depend on the metallic and not on the metall*oid* element which they contain.

We must finish this hasty notice with a *résumé* of M. Rabuteau’s ingenious theory as to the cause of the constipation which so often succeeds saline purgation. According to him, if the dose of the saline be large almost the whole is directly eliminated by the alimentary canal; if it be medium, a part passes into the blood; if it be small, nearly the whole of it is absorbed into the blood. In the first case, purgation is violent; in the second it is slight; in the third it is *nil*, and there is even constipation. But as a certain quantity of the medicine may have been absorbed, even when there has been powerful purgation, there may be consecutive constipation from the presence in the organism of the purgative salt, which slowly eliminates itself from the intestinal surface. Rabuteau inclines to think, though it is impossible to affirm, that not merely saline, but other purgatives, show analogous differences of action.

In concluding our notice of this *Annuaire*, useful and interesting as it is, we cannot but repeat the complaint we made last year. It is astonishing to what an extent the accomplished editor has ignored some of the most valuable therapeutic work done both in Germany and in England. We venture to say that he would have done better to attend to these matters, than to fill so many pages as he has done with a reproduction of his own papers on the etiology of saccharine urine, which is clearly beyond the proper work of his *Annuaire*.

De la Médication Antipyrétique. Thèse de Concours. Par le Dr. A. Ferrand, Ancien Interne Lauréat des Concours des Hôpitaux, &c. &c. Paris: F. Savy, 1869. 8vo. pp. 90.

(*On Antifebrile Medication.* By Dr. A. Ferrand.)

The author of this treatise is a really distinguished man in the *jeune médecine* of France; and it is with a natural interest that we turn somewhat eagerly to the pages of his thesis to discover what are his ideas as to the part which treatment can play in pyrexia. One can hardly do better, in reviewing his work, than select the chapter on the “Indications in Fever,” as a kind of test object, to judge the quality of his work. Judged by this test, it must be pronounced very good. In a quiet and thoughtful manner, M. Ferrand inquires into some of the deepest problems of the physiology and chemistry of fever, and really hits, as it seems to us, most if not all of the principal difficulties which are troubling the minds of the most advanced pathologists and clinical observers in Europe. That he does not pretend to solve all these mysteries is no dispraise to him, but the reverse: at any rate it may be fairly said that he has carefully considered all the doubtful points by the light of the best observations in nearly every European country. If we were inclined to make any exception to his accuracy and completeness of information, it would be on the score of what he says, or rather omits to say, respecting the *rôle* played by hydro-carbonaceous foods in alimentation. He seems to us to greatly undervalue, if not to ignore, the incalculably important results of recent researches in deciding the rank of non-azotised aliments in *feeding*, and consequently in great part *disarming*, the destructive force of pyrexial combustion. He assigns, as it appears to us, an altogether exaggerated importance to the secondary effects of pyrexia upon the nervous system: while at the same time he appears inadequately impressed with the enormous destructive incidence of febrile action upon the tissues. Upon this point M. Ferrand would surely do well to consult the description of what passes in the organism in pyrexia which was given by Professor Haughton in his admirable address before the British Medical Association at Oxford. He seems to forget, what Professor Haughton therein so ably showed, that a typhus fever or pneumonia patient lying still on his back, and with scarcely anything moving except the organs of vegetative life, and the deep chemistry of the tissues, does in fact a heavier day's work than any healthy labourer!—heavier, that is, as regards the inevitable destruction of tissue that must go to the maintenance of the most elementary and necessary acts of life, in the absence of the power to assimilate ordinary nutriment. It is this defect (as we think it) in his physiology which makes M. Ferrand's practical remarks on alcohol so very inferior to those which he makes, in the practical therapeutic portion of his work, upon other agents which he thinks appropriate to the treatment of the febrile state. His account of Todd's doctrine and school is indeed extremely inadequate, and proves, for the hundredth time, how much that remarkable teacher has been misunderstood by the majority of those who have criticised his opinions, or supposed opinions. Here is a sentence, for instance, of M. Ferrand's, which is nearly as incorrect as it is possible for a sentence to be—"Alcohol, for instance, is ... stimulant and resolvent in small doses; but in larger and more continued doses it becomes antipyretic." That is precisely what alcohol does not do. Given in large doses (relatively to the needs of the organism), it becomes eminently pyretic. It need not always raise the temperature of those parts to which we apply the thermometer; but, assuredly, given in such doses as produce phenomena of intoxication, it does most directly increase and give a mischievous impulse to the destructive processes going on in the organism.

We have no wish, however, to leave an unfavourable impression of M. Ferrand's very able pamphlet on our readers' minds. On the contrary, we refer them with confidence to the work as a repository of a large amount of accurate and careful thought and observation on the nature and the remedies of the pyrexial state.

A Practical Treatise on Perimetritis and Parametritis. By J. MATTHEWS DUNCAN. Edinburgh: Adam and Charles Black. 1869.

Though we have nothing to do with the pathology of this work, it may be as well to explain the meaning of the terms employed in the title, so that the value of the author's therapeutics may be the better understood. Objecting to such expressions as pelvic cellulitis and inflammation of the uterine appendages, Dr. Duncan adopts in part the phraseology of Virchow, and employs the words perimetritis and parametritis, the former to signify inflammation of the uterine peritoneum, and the latter to imply inflammation of the cellular tissue in connexion with the uterus. With the justification of such a terminology we need not concern ourselves, but we may express a regret that so much of the author's observations are confined to the natural history of these affections, and so little to the all-important problem of treatment. In a work extending over nearly 250 pages, one expects to find therapeutics represented by a greater space than that included in about a sheet of printed matter. Our disappointment, too, is enhanced when we find the author, in many instances, limiting himself to the vaguest of generalities, and, while sceptical as to opinions which do not coincide with his own ideas, credulous to a high degree on some points of traditional medicine. The only methods of treatment on which Dr. Duncan at all dwells are those of leeching and poulticing—save that, in a few words on internal remedies, he urges the use of mercury to produce slight salivation, and rejects, what many think so valuable, the employment of opium. He is strong on the subject of poulticing, and his statements are in accordance both with practical experience and *à priori* reasoning. He impresses seriously on his readers the importance, during the acute stage, of keeping up the poultices constantly night and day. As to blood-letting, his practice is definite, though his arguments from physiology in support of it are, we must confess, insufficient for us. Local leeching is, in his opinion, vastly superior in its effects to more distant venesection; and doubtless there is much good in the practice of applying a few leeches over the groin or to the perineum. But the following argument in favour of the former plan of distant blood-letting strikes us as being of that painfully unprecise character which unhappily is so much associated with medical research:—

“The profession in this country at least has lost all faith in this treatment, as well as in the corresponding doctrine regarding venesection of special veins of the upper extremity in disorders of the head. But enough remains in the well-known and, it appears to me, well-founded belief in the value and efficacy of the pediluvium in menstrual affections to prevent us from regarding these therapeutics as absurd; and, although not dreamt of in our modern and too self-sufficient medical philosophy, yet laws of sympathy between distant parts may be discovered which will explain and inculcate some such remedial measure, which now appears to be unreasonable.”

How, in the name of all that is “positive” in medical science, is therapeutics to advance an inch while philosophers reason to truth in this fashion? It would not be more absurd for a chemist to support a gratuitous speculation on the faith of a future recognition of phlogiston, than for an intelligent practitioner to establish a therapeutical fact by argument such as that which Dr. Duncan employs. When the author confines himself to telling us under what circumstances blood-letting should be adopted, he gives us the result of a valuable and wide clinical experience; but his hypotheses are, we confess, too much for us. We cannot understand why Dr. Duncan completely overlooks the subject of restoratives and tonics in perimetritis and parametritis; and we should be glad to hear his reason for ignoring such very important agents in the treatment of these affections.

The Atlas of Venereal Diseases. By M. A. Cullerier. Translated from the French, by F. J. Bumstead, M.D. Philadelphia: Henry C. Lea, 1868.

We wish for once that our province was not restricted to methods of treatment, in order that we might say something of the exquisite coloured plates in this fine volume; for the work is essentially one to aid in diagnosis rather than to detail means of cure. The Atlas, which Dr. Bumstead has not only translated, but very materially added to from his own stores of knowledge, is in every respect a most useful work for the practitioner, who is often called on to diagnose an affection which in the absence of a truthful history may appear either syphilitic or not in nature. With the aid of these handsome plates, there need be little difficulty in the identification of a syphilide. It must not be supposed, however, that therapeutics are neglected, or sacrificed to etiology. Both author and editor give us a very full account of the remedies now in vogue, and of their own clinical observations. We have not seen anything on the subject of the hypodermic employment of mercury, but the internal administration of the salts of mercury and iodide of potassium is of course enjoined. Indeed, the chapters on the treatment of syphilis are not the best. The section devoted to the remedial measures to be attempted in gonorrhœa strikes us as being copious and well arranged, and contains some sound, practical commentaries by the editor, who disapproves of the porte-caustique and other heroic modes, and recommends the use of an extremely weak injection of nitrate of silver (gr. ⅙ to the ℥j.) every three or four hours. His suggestions as to general treatment are equally judicious. In every respect this Atlas will be found most useful for reference by the busy practitioner.

The Medical Formulary, &c. By B. Ellis. Twelfth Edition, revised by Albert H. Smith, M.D. Philadelphia: Lea, 1868.

The aim of this work is to supply the young physician with the means of writing “elegant and judicious” prescriptions; and if we may judge by its success, the book must be one which meets a want. But we cannot help saying that the habit of writing “elegant and judicious” prescriptions is one of the barbarisms of the practice of ancient times which we should gladly see consigned to oblivion. It fosters charlatanism, and utterly retards all efforts to found a rational system of therapeutics. How can any logical induction, or any generalization of the slightest value be drawn as to the remedial effect of drugs administered after the mode laid down in such books as the “Formulary?” Without pausing to consider a recipe for pills which are “elegantly” and alliteratively styled “Chapman’s peristaltic persuader,” let us ask what can be the judiciousness of the following marvellous concoction of substances?—

Elixir of Cinchona.

℞ Quiniæ sulphatis, gr. xxv.
Quinidiæ sulphatis,
Cinchoniæ sulphatis, āā gr. x.
Sacchari, ℥xx.
Olei anisi,
Olei fœniculi, āā gtt. ij.
Olei cinnamonii Zeylandici, gtt. vj.
Olei cari, gtt. j.
Olei aurantii, ♏xl.
Spiritus Curaçoæ, f. ℥vj.
Alcoholis deodorati,
Aquæ rosæ,
Aquæ, āā Oj.
Caramel, ʒiij.
Misce secundem artem!

Conservative Surgery, with Reports of Cases. By ALBERT G. WALTER, M.D. Pittsburg: Johnston.

This is a very verbose treatise on the mode of dealing with lacerated wounds, &c. The author recommends free incisions along the whole length of the limb, and the subsequent application of poultices and fomentations, assisted by general and local supporting measures. As is usual in such cases, he cites a vast series of cases, which, as is equally usual, might be cited in evidence of a very large number of different and conflicting propositions. There is some sense in the author’s practice, but a terrible deal of nonsense in certain of his physiological speculations.

Clinic of the Month.

The Pressure and Ligature Methods of treating Aneurism.—In the course of a lecture on aneurism of the femoral artery Mr. Paget, adverting to these two methods, speaks of them thus: “Taking large numbers of cases of aneurism together, they are very nearly balanced, on the one side for pressure, and on the other for ligature. In favour of pressure there is the experience of the surgeons in Dublin. They seem to have a much larger number of aneurisms, especially of aneurisms of the popliteal artery, to treat than we have in England, and they have certainly a large amount of success. I have no doubt this is in part due to a well-arranged system, and to the house-surgeons and dressers acquiring a more special skill than we have yet achieved. On the other hand, there is the remarkable success attained by surgeons who have constantly practised, with great skill, the ligature. The success of Mr. Syme in the ligature of the femoral artery for popliteal aneurism has been so great that any one who might fairly expect to attain nearly the same measure of skill would undoubtedly follow the ligature rather than the pressure. I would prefer, however, to leave the subject open for your own observation, and say, endeavour to ascertain, as far as you may be able, which are the cases for the ligature and which are those in which pressure is more likely to lead to a good result. And in many cases in respect of which you are doubtful, pressure may be tried first and the ligature afterwards.” (See *Lancet*, April 24.)

Treatment of Atonic Dyspepsia.—Dr. Thorowgood, in a paper just published, refers to the existence of torpor of the colon as a complication in cases of this kind. In treating cases of dyspepsia occurring in those who work hard with their brains and have but little “tone” about the stomach and bowels, he is convinced that the more we refrain from the administration of purgatives the better. At one time he used to think that when the tongue was crusted a purgative could not be amiss; but to whatever degree this holds good with strong country people and over-fed townspeople, it does not apply to those who have feeble appetite and who work hard. In these cases he has seen an acid mixture or a chalybeate do more service in cleaning the tongue and promoting digestion than alkalies or aperients. If the colon be filled with scybalæ, the best evacuant is a table-spoonful of castor-oil in peppermint water. When the constipation takes on a less or more obstinate character, he uses a saline chalybeate in imitation of the saline chalybeate waters of Kissingen, Harrogate, &c. In addition he gives a zinc pill, with extract of henbane, at night. But he avoids the use of opium. (See *Lancet*, April 24.)

Anaemia and Chlorosis treated by Nickel and Manganese.—Dr. Broadbent lately read a paper before the Clinical Society (April 9th), in which, on the principle that chemical substances closely allied have similar action, he recommended manganese and nickel as substitutes for iron in the treatment of chorea. He recorded various cases, in some of which good results appeared to follow this method.

Operation for Chronic Inversion of Uterus.—At the meeting of the Royal Medical and Chirurgical Society, on the 13th of April, Dr. Barnes read a paper, in which he described a new operation for the relief of chronic inversion of the uterus. He gave the statistics of the different methods now in use. He stated that the ligature and excision were open to the double objection that, besides being very hazardous to life, success was only achieved at the expense of mutilating the patient. Forcible taxis was a violent and often fatal proceeding. Sustained elastic pressure had given remarkable results, but cases would occur where the constricted cervix uteri would resist simple pressure. He then described a case of inversion of six months' standing, which had resisted elastic pressure kept up for five days, and in which he resorted to a plan then practised, he believed, for the first time, of making three longitudinal incisions into the os uteri, so as to relax the circular fibres; taxis then applied quickly succeeded, and the woman made an excellent recovery. He proposed, therefore, as the best proceeding where simple sustained elastic pressure fails, to make an incision on either side of the os uteri, and then to re-apply the elastic pressure, as being safer from the risk of laceration than the taxis. (See *British Medical Journal*, April 24.)

A Presse-artère for Compression of the Arteries, which may be found useful in some cases, has been described by Mr. B. Wills Richardson, of Dublin. The tubular *presse-artère* which he has invented is intended only for immediate compression, but it was used with good result in the amputation of the fore-arm. The new instrument is composed of two parts. (1) A fine silver or a fine German silver tube. To the upper end of this tube a small milled button is soldered. The button facilitates the turning or screwing of the tube by the fingers of the surgeon. A female screw is formed upon the upper half of the inside of the tube. (2) A steel stem having two jaws at its lower end. These jaws are perfectly smooth on their opposed as well as on their outer surfaces, and free from any cutting edge. They are so arranged as to open and close parallel to each other. At the upper end of the stem there is a handle nut. It is hexagonal only in the present instrument; but, for recognition in wounds, the handle nut of the *presse-artère* intended for large arteries should have some other form when more than one instrument is in use. The nut is fitted to the stem by means of a square mortise to prevent it from turning on the stem during the screwing or unscrewing of the tube. The handle nut is secured in its position by a smaller but screw nut. The upper half of the stem has a male screw cut upon it, and is adapted to the female screw on the inside of the tube. The inventor claims the following advantages for this piece of apparatus:—(1) The smallness of the space it occupies. (2) Facility of application and removal. (3) Accurate graduation of the compression by means of the fine screw arrangement. (See *Medical Times*, April 24.)

Catgut in the Ligature of Arteries.—In a communication to the *Lancet*, Professor Lister, after giving numerous pathological details, refers to the practical importance of catgut as a ligature. He states that by applying a ligature of animal tissue antiseptically upon an artery, whether tightly or gently, we virtually surround it with a ring of living tissue and strengthen the vessel where we obstruct it. This antiseptic animal ligature consists of catgut steeped in carbolic acid and oil. And with such a ligature Professor

Lister says he should now "without hesitation undertake ligature of the innominate, believing it to be a very safe proceeding." He thus expresses himself as to the necessary qualities of the ligature:—"The method which I have found to answer best is to keep the gut steeped in a solution of carbolic acid in five parts of olive oil, with a very small quantity of water diffused through it." A larger proportion of the acid would impair the tenacity of the thread. If a mere oily solution is employed, the gut remains rigid, the oil not entering at all into its substance. But a very small quantity of water, such as the acid enables the oil to dissolve, renders the gut supple without making it materially weaker or thicker. And, curiously enough, the presence of this small amount of water in the oily solution gradually brings about a change in the gut, indicated by a deep brown colour; after which it may be placed in a watery solution for a long time without swelling, as a portion prepared in a simple oily solution does. This is a great convenience; for an oily solution is unpleasant to work with during an operation, and exposure to the air soon renders gut suppled with water rigid from drying. But when it has been treated in the way above recommended, it may be transferred to a watery solution at the commencement of an operation, and so kept supple without having its strength or thickness altered. "For tying an arterial trunk in its continuity, catgut as thick when dry as ordinary purse-silk will be found best. But for ordinary wounds, where, if one ligature happens to break, another can be easily applied, much finer kinds may be employed, and are convenient from their smaller bulk." (See *Lancet*, April 3.)

The Advantages of Tracheotomy.—Mr. T. R. Jessop corroborates the views expressed in our last Number by Mr. A. E. Durham. He gives some very remarkable cases, showing the great usefulness of the operation. Indeed, one case was a veritable resuscitation of life. (Ibid.)

Arsenic a Cause of Shingles.—The very important problem as to whether arsenic, when continuously administered, is productive of shingles, is again discussed by Mr. Hutchinson. Mr. Hutchinson does not assert the fact to be more than a coincidence, but he relates several very interesting cases in which the prolonged use of arsenic was followed by shingles. (See *Medical Times*, April 17.)

The Treatment of Diabetes.—Dr. Basham records some cases of diabetes treated very successfully with alkalies and the phosphatic salts of ammonia, and he expresses an opinion very favourable to this method of treatment. The following is the prescription employed:—Phosphate of ammonia and carbonate of ammonia of each ten grains, aromatic spirit of ammonia half a drachm, water an ounce, added to the juice of a fresh lemon, and taken three times a day. This line of treatment was continued for four months, with the results tabulated below:—

	Mean sp. gr.	**Sugar per oz.**
September (began phosphatic salts)	1037	18 grains.
October	1040	18 grains.
November (great increase of urates)	1036	6 grains.
December 4th (large proportion of urea and urates)	1018	½ grain.
December 28th (urea and urates in excess, a large crop of crystals of oxalate of lime after cooling)	1024	Nil.
January 26, 1869 (same as above)	1026	Nil.

(See *British Medical Journal*, April 10.)

Silver-wire Ligatures for the Pedicle in Ovariotomy.—Dr. Marion Sims calls attention to a recent case in which he performed ovariotomy, in order to explain his method of dealing with the pedicle. In the particular instance the walls of the abdomen were so thick, and the pedicle was so short, that it would have been impossible to secure it in the usual way by the clamp externally. In such cases he has always held that the pedicle is best secured with silver-wire ligatures and dropped back into the pelvic cavity. "If the pedicle be small, it is enough to transfix it with a double silver-wire and secure the two halves by firmly twisting the wires on opposite sides." If it be broad, it requires a number of separate wires. This case required eight deep silver sutures for closing the external wound, care being taken to pass them through the divided edges of the peritoneal coat. A piece of lint, wet with carbolic lotion, was laid over the wound, and secured by a bandage, and a large compress of cotton wadding. The urine was drawn off for three days. There was no constitutional disturbance, and the patient was convalescent from the moment of the operation. (See *British Medical Journal*, April 10.)

Perchloride of Iron in Post-partum Hæmorrhage.—Mr. Hugh Norris corroborates the testimony of the various obstetricians who have spoken so favourably of this styptic. Injections of strong solution of the salt instantly arrest the hæmorrhage in this dangerous class of cases. He noticed also that the perchloride has a peculiar corrugating effect on the superficial muscular fibres, as well as on the mucous surfaces. He has noticed in less than five minutes after injection that the sphincter vaginæ, which had previously allowed the passage of the hand, became so contracted that it barely admitted a single finger. From this he concludes that one of its beneficial effects in these cases is the contraction of the muscular fibres of the uterus. He lays down the following conclusions in reference to this preparation:—(1) We possess no topical styptic in efficacy at all approaching the perchloride of iron; its effects being certain, perfect, and instantaneous. (2) In *post-partum* hæmorrhages, a solution of this salt, applied in the form of an intrauterine injection, is of the utmost value both in immediately arresting the flow of blood and also in causing a permanent contraction of the recently emptied uterus. Its presence in the cavity of the uterus *post-partum* is not only not injurious, but on the contrary, from its well-known antiseptic properties, may frequently be productive of positive benefit in more ways than one. Mr. Norris employs a saturated (?) solution of the perchloride. All clots must first be removed, and the long tube of the syringe should be introduced thoroughly into the uterus before injection. (Ibid.)

The Cure and Prevention of Scurvy.—In a paper in the *Lancet*, Mr. Archer Farr starts a doctrine which is yet, we think, to be proved, viz. that scurvy is not caused by the absence of certain alkalies and the presence of others. Indeed, he refers scurvy to the absence on ship-board of proper flesh-food, and he thinks that by supplying flesh in thoroughly good condition scurvy may be avoided. Lime-juice acts, he says, by taking the place of the gastric juice and digesting the food, and thus promoting the nutrition of the body (!). (See *Lancet*, March 27.)

Treatment of the Vomiting of Pregnancy.—Mr. John Harrisson recommends that in these cases hypodermic injection of morphia be tried. He gives the report of a very decided and serious case, in which nearly every conceivable remedy had been employed in vain. He then tried the subcutaneous injection of acetate of morphia, in doses of one-sixth of a grain, three times a day, and this instantly arrested the vomiting. (See *British Medical Journal*, April 3.)

Extracts from British and Foreign Journals.

Carbolate of Soda as a Remedy for Itch.—Dr. Zimmermann, of Braunfels, remarks that no one who sees much of itch will deny that we are without any remedy which acts with the certainty of a specific. In private practice, where we cannot readily obtain the proper baths, frictional manipulations, &c., cases are apt to be very inveterate. The popularity of petroleum and Peru balsam is due chiefly to their being neither very disagreeable nor very troublesome in the use; but petroleum has not justified its reputation,[12] while Peru balsam, which really is very valuable, especially in recent and in children's cases, is unfortunately very costly. Zimmermann is inclined to hope that in carbolate of soda he has found a remedy that will cure scabies, *tuto*, *cito et jucunde*, though his experience is not yet sufficient for absolute proof. He employs a solution of 160 to 320 grains of the salt in about 7 ounces of water; this is to be well rubbed into the affected parts thrice daily. In two or three days every case of Zimmermann's, even the inveterate ones, has been completely cured, and this without any annoyance or interruption of the patient's ordinary business. There is no irritative erythema of any consequence from the frictions. Carbolate of soda may be used as a disinfectant and deodoriser; for this purpose 16 to 32 grains to 7 ounces of water is sufficient. (*Der Praktische Arzt. März.*)

On the Contra-indications of Anæsthesia.—M. Gosselin considers that alcoholism renders patients very unfit for taking chloroform, and thinks that to all persons above 50 years of age, and given to intemperance, chloroformisation should be forbidden; or at any rate only applied with the greatest caution, and never for a long time together. Professor Nagel, of Vienna, has recently opposed this wholesale condemnation. In delirium tremens, chloroform narcosis is often very useful, especially where it is necessary to set fractures. The severest delirium, on which large doses of opium produce no effect, has been repeatedly calmed by chloroformisation in Nagel’s own practice. Alcoholism only so far contra-indicates anæsthetics, that in refractory subjects it is necessary to push the agent in such large doses that, even with the greatest care, there is a risk of asphyxia.

An even more positive contra-indication to chloroform is found by Gosselin in the case of stupor following severe wounds. No one will dispute that narcotisation is at first entirely out of the question; it only remains doubtful for how long a time after the injury it is unsafe to give chloroform for operative purposes. Gosselin is for total exclusion, since even after the apparent departure of stupor, a kind of concealed shock to the system may still exist, and would render anæsthesia dangerous. He also thinks that in recent dislocations, especially of the shoulder, chloroform is not only usually needless, but contra-indicated by the fact that the patient’s want of sensibility may permit such force to be used as may inflict severe injuries upon the nerve trunks; and even in old dislocations chloroform should not be used till other means have been tried. This contra-indication, also, is opposed by Nagel, who brings a large amount of experience to controvert it; it even happens, sometimes, that the muscular tone can be sufficiently relaxed for the reduction without any loss of consciousness. The chief contra-indication to chloroformisation which Nagel admits is the pre-occurrence of long-standing or considerable (arterial) hæmorrhage, and advanced age, especially if there be also heart or lung disease, vascular degeneration, emphysema, &c., also the fact that the operation might cause blood to enter the larynx; and in hernia, because of the tendency to vomit. He recommends the greatest care in giving it to refractory patients, who struggle, scream, and hold their breath. The danger of asphyxia increases every moment, and it is necessary to have ready the means of throwing in a stream of pure air or of oxygen, through the *nose*, sprinkling the patient with cold water at the same time. Nagel has also observed that the restlessness and oppression produced by half-felt pain in incomplete narcosis, and the consequently insufficient respirations, may produce an even fatal exhaustion. Narcosis should therefore be properly kept up as long as it is wanted. (*Der Praktische Arzt.*)

[It is very necessary that we should study the opinions of foreign physicians and surgeons as to the dangers of anæsthesia. At the same time, I think it right to protest here that the cautions given even by Nagel, and still more those which Gosselin inculcates (with two exceptions), are to me inexplicable. At least they can only be understood upon the supposition that the means of inducing anæsthesia which are employed by these authorities are extremely inefficient and improper. A former very large experience of chloroform administration, some years ago, impressed me with the confident belief that if chloroform be given with proper care and with a Snow's, or still better a Clover's apparatus, there is really no danger whatever in its use, except in two cases—that of shock after severe injury, and that of delirium tremens. I cannot admit that a patient who is in a fit condition to undergo a surgical operation at all is placed in any worse position for supporting the shock of it by the fact that he has been chloroformised, if this has been skilfully done—the case of severe shock from injury always excepted.—F. E. ANSTIE.]

Subcutaneous Injection of Mercury in Syphilis.—Dr. A. Stöhr records a series of clinical observations on this method of treatment, which are very interesting. He considers that the subcutaneous injection of the bichloride is the most effective and direct means of producing the curative effects of mercury which has ever been applied. Stöhr employed the treatment for 96 patients; in almost all the cases it was carefully ascertained that no specific treatment had been previously adopted. His cases included a larger relative proportion of inveterate syphilis than those treated by Lewin in his researches, described by M. Bricheteau in the *Practitioner* for March. Stöhr considers that the hypodermic method is not needed for the milder forms of syphilis. Old and obstinate cases, on the other hand, in which inunction or other forms of mercurialisation have been vainly tried, are particularly appropriate for a trial of the subcutaneous method. It is especially indicated in such cases as those of iritis and of dangerous laryngeal affections, in which it is important to produce a very rapid effect; also in severe and extensive syphilitic skin affections, in which inunction cannot be applied. Syphilis of the bones and periosteum, and syphilitic gummata, are but little influenced by hypodermic mercurialisation. In cases accompanied by severe marasmus, the hypodermic use of mercury is contra-indicated. It is also inappropriate to the treatment of those persons who have to be treated as *out*patients of a hospital. Stöhr employs a solution of corrosive sublimate in distilled water: he had at first tried a solution in glycerine, but this did not prove practically convenient. The dose ordinarily used was ⅛ of a grain; the daily employment of this only slowly produces ptyalism. When ¼ grain doses were used daily, the slighter symptoms of ptyalism never failed to occur by the third or fourth day, and the severer phenomena by the eighth or ninth; so that, with few exceptions, the administration of 2 or 2½ grains in this way produced such a strong development of salivation that the treatment had to be interrupted. (*Deutsches Archiv f. klin. Med.* V. 3 and 4.)

The Treatment of Diabetes.—Dr. Leube gives an elaborate report of two cases of diabetes in which he made the most careful daily observations of the quantity of water, of sugar, &c. He arrives at the following therapeutic conclusions:—Pure meat diet (with only *almond* bread) was the most powerful means of reducing the sugar excretion. Of drugs which were tried, *arsenic* had by far the most remarkable effect in reducing the sugar. Saikowsky discovered, some three years ago, that the continuous administration of arsenic for several days to animals entirely removed all glycogen from the liver; and that then neither puncture of the fourth ventricle nor curara poisoning would produce diabetes at all. Leube made the therapeutical application of the drug which these experiments suggest. He administered Fowler's solution in doses equivalent to about ⅓ grain of arsenic daily. The effects were most striking during the period when the patients were taking a *mixed* diet. With mixed diet, and without arsenic, the daily average of sugar was 570 grammes in one case; arsenic reduced it to 352 grammes, on the average of 79 day and night observations; and substantially the same result was obtained in the other case. The use of this drug would appear to promise results of real importance. (*Deutsches Archiv f. klin. Med.* V. 3 and 4.)

A German Criticism of Lister's Treatment of Abscesses.—Dr. W. Rosco publishes a very sharp critique on this plan of treatment, which has lately become so fashionable. He analyses minutely the sixteen cases which were published in the *Archiv f. Heilkunde* (and reported in the *Practitioner* for July 1868), and maintains that the results obtained were not more favourable, if so favourable, as those which are often obtained without any use of carbolic acid. Rosco maintains that there are two weighty objections to the too free and indiscriminating use of this treatment for abscesses. In the first place, there is a danger that surgeons, trusting blindly to the antiseptic action of the carbolic acid dressing, will open cold abscesses either unnecessarily early, or even in cases where incision is altogether improper. Secondly, the caustic or irritant action of the acid will occasionally produce mischievous effects. But the main point of his argument is directed to the demonstration, that even the most brilliant results which have been obtained in England, and published by Lister and others in the columns of the *Lancet*, are by no means conclusive in showing that the carbolic acid was really the curative agent. He observes, moreover, that although in England the treatment has been most extensively tried everywhere, the majority of hospital surgeons appear very sceptical about it. Undoubtedly Lister's method deserves every attention, and should be tried in appropriate cases, but the inquiry should not be made with too much credulity, but with prudent doubt. (*Archiv f. Heilkunde*, 2, 1869.)

Electricity in the Diseases of Children.—Dr. Ullersperger gives a good summary of existing knowledge as to the uses of electricity in the paralyses of common and special sensation, and of motion, in children. The paper is a useful one, but does not contain any original matter on which it is necessary to comment here. (*Journ. of Kinderkrankheiten*, and Jan. Feb. 1869.)

Liebig's Food for Infants.—Dr. Kjelberg related to the *Gesellschaft schwedischer Aerzte* his experience of the use of Liebig's food for infants as a remedy. Six cases of diarrhœa occurred in the Children's Hospital among infants of from 1½ to 2 years; five of them had already been treated with medicine without effect. A thin broth made from the "food" was given them as their only nourishment, and all medicine was discontinued. The motions at once assumed a better appearance. In one case, which had no previous treatment, the effect of the exclusive use of Liebig's food was very striking. Kjelberg says that he had used the treatment in two cases of children, private patients, in whom not diarrhœa, but obstinate constipation was the malady. The children were still suckled, while the food was administered. The peristaltic function of the bowels rapidly became normal and regular. Kjelberg thinks that Liebig's food possesses the capacity of *regulating* the activity of the intestinal canal. (*Ibid.*)

Hypodermic Injection of Ergotin in the Treatment of Aneurism.—Professor Langenbeck writes an important paper on this subject. From the well-known influence of ergot in provoking contractions of the organic muscular film of the uterus, he was led to think that a similar stimulation might be produced by it, with beneficial effect, in the muscular coats of arteries in cases of aneurism. The first case in which he employed it was one of subclavian aneurism in a man aged 45. The tumour was treated in the first place on Jacobson's plan, with the repeated application of moxas, and a great diminution of all the symptoms took place, and lasted for three years; but the pulsation continued. In consequence (as the patient thought) of an excessive summer heat, a relapse took place, the tumour enlarged greatly, and all the old symptoms returned. Three centigrammes of aqueous extract of secale were injected over the tumour, with great relief to the pain and consequent insomnia. Between January 6 and February 17 about 30 grains of ergotin were injected, with the effect of so greatly relieving the symptoms of pressure in veins, that the pain and paralysis of the arm and hand were diminished to a remarkable extent. The pulsations of the aneurism were also sensibly weakened, and the tumour somewhat sunken. In a second case—one of aneurism of the radial, about an inch and a quarter above the wrist, and which had existed for many years—about 1/4th of a grain of aqueous extract of secale (dissolved in seven times its bulk of half and half glycerine and sp. rectif.) was injected into the skin above the tumour, and on the following day the tumour appeared to have vanished. The cure became permanent, and the only trouble was a local erythematous inflammation which lasted some days. Langenbeck discovers that it was natural that a more powerful effect should be produced by the remedy in radial than in subclavian aneurism, since the radial artery is more copiously furnished with muscular fibres than the subclavian. (*Berlin Klin. Wachensch.* 12, 1869.)

Pruritus of the Skin of the External Auditory Meatus.—Dr. J. Gueber describes under this name, not the itching and eczema of the ear so often left after scabies, but a special affection, first named as above by Hebra, in which the itching is the *only* symptom. It occurs most frequently in middle age, and in persons who, from one reason or another, have a defective circulation. There are frequently periodical exacerbations of the irritation.

Gueber recommends, as a palliative, applications of water and oily matters for radical cure, repeated painting with a strong solution of nitrate of silver, till inflammatory reaction is set up. Along with this, however, there must of course be a suitable rational treatment, according to the special indications which present themselves. (*Allgem. Wien. Med. Zeitg.* 52, 1869.)

New Remedies.—Dr. G. W. Lawrence, writing to the *Philadelphia Medical and Surgical Reporter*, states that the following substances, whose qualities he briefly describes, have been recently tried by him in practice:—*Quiniæ Iodo-Sulphas*. It has proved to be a desirable alterative and tonic, serviceable in consecutive syphilis, scrofula, cachexia, neuralgia, some forms of paralysis, and in debilitated conditions of the general system. Given in pill form, or mixed with syrup of sassafras bark or blended with elixir of Calisaya bark.—*Iodide of Antimony.* An alterative, used chiefly in skin diseases, those forms arising from constitutional ills and secretory disturbances. Prescribed with aromatic syrup of dulcamara. He has recently used iodide of antimony as an anaphrodisiac with satisfactory results. He combined it with lupulin, and gave it in pills every six hours.—*Iodide of Manganese.* Alterative, administered generally with the iodide, or some other desirable preparation of iron, or with quiniæ iodo-sulphas, in anæmia and chloro-anæmia, with resin podophyllin, in chronic splenic and hepatic derangements. In the alterative agency of manganese, he fancied that a determined action is exercised on the ganglionic system of nerves.—*Glycerole Pyrophosphate of Iron.* In the formula pure glycerine is substituted for sugar, or simple syrup; each fluid ounce (with glycerine) contains sixteen grains of pyro-phosphate of iron. This new preparation is unchangeable, and is one of the most palatable of that family of tonics. He employs it usually as an eutrophic in that spanæmic condition of the system so frequently provoked by the protracted use and abuse of iodides and bromides of potassium. He also uses it when indicated, in progressive paralysis, motor ataxia, in threatening a supposed incipient *ramollissement* of the brain and spinal marrow.

Carbolate of Lime in Pertussis.—Dr. Snow, of Providence, has suggested the use of carbolate of lime in hooping-cough, and in all cases it has apparently produced a marked effect in diminishing the frequency and severity of the paroxysms. Small quantities of the carbolate of lime are placed in saucers in the room where the child sleeps; merely sufficient to make the odour perceptible. (*New York Medical Record.*)

Bromide of Potassium and Antimony in Puerperal Convulsions.—Dr. T. N. Simmons reports the history of a case of puerperal convulsions, in which the efficacy of these remedies was evident. A primipara, while in labour, with the head of the child in the inferior strait, was seized with a violent convulsion, which was followed by four others, with an interval of about 15 minutes between each. Chloroform proving of no benefit, bromide of potassium was administered, beginning with a dose of 40 grains in combination with half a grain of antimony. In combination with the bromide one-half grain of the antimony was given every hour and a half or two hours, until three grains of the antimony were taken. After the first dose there was a return of four paroxysms. The first occurred within an hour, the second in two hours, the third between three and four hours, and the fourth in eight hours. Their intensity and duration were also diminished in the order of their recurrence. Convalescence was rapid. (*N. O. Journal of Medicine.*)

An Acidulated Solution of Pepsine as a Solvent for False Membrane in Diphtheria.—Dr. W. H. Doughty has communicated to the *Richmond and Louisville Medical Journal* an article upon this subject, with a history of a case in which he was entirely satisfied with the efficacy of pepsine in diphtheria. The patient was about 25 years of age, of feeble general health from intermittent fever. He presented himself with sore throat. For the affection cauterization was resorted to, and a gargle of chlorate of potash ordered. The throat became very much inflamed and swollen, and the glands about the neck enlarged. About the fourth day exudation of membrane was observed under the tongue, a portion of which was removed with forceps. The swelling increased, and the membrane continued to re-form. Quinine and stimulants were freely used, with inhalations of lime-water. On the fifth day commenced the application of pepsine to the membrane, keeping up the same general treatment. Pepsine was used in the following proportion: ℞ pepsine ʒj; acid muriatic, dilute, gtt. x; water q. s. ad ℥iij.: M. and filter. This was applied by means of a hair pencil continuously. A few hours from the commencement of the application, "the mouth, as far as visible, is cleaner and better." On the next day the patient feels better; no appearance of exudation; mouth is clean, but continues to discharge broken-down opaque masses from the throat, and thinks he must have suffocated but for the solution employed. The breathing is comparatively easy and cough less. Patient died on the seventh day, of asthma.

Impermeable Dressings in Eczema.—In an article on this subject in Mr. Erasmus Wilson's "Journal," Mr. Alfred Pullar makes the following remarks on the value of this mode of dressing:—"The method of local treatment first brought into notice by Professor Hardy at the Saint Louis Hospital in Paris, consists in covering the diseased parts completely with vulcanized india-rubber cloth (*toile caoutchouquée*). The material used for this purpose is ordinary cotton cloth covered with a solution of caoutchouc and subsequently vulcanised: by this means it is rendered impermeable to watery fluids, and acquires on one side a smooth surface. The therapeutical effects resulting from a covering such as that described would seem to depend essentially upon two conditions:—*First*, the exclusion of the air. As it has been proved by experience that the influence of the atmosphere increases the inflammation of the diseased surface, its complete exclusion fulfils an important indication in the treatment: this is accomplished by the india-rubber covering, which also protects the abnormally sensitive skin from variations of temperature. *Second*, the retention of the secretions of the skin. These—exuded in considerable quantity, and *unchanged* by the atmosphere—are retained in contact with the skin, and seem to act by relaxing the inflamed structures. Whilst visiting Hardy's wards at the Saint Louis Hospital, I had the privilege of seeing several cases of eczema treated by this means (these cases being chiefly eczema of the limbs in old people). The impermeable dressing was so applied as to cover completely the affected parts, and was removed, from time to time, in order to be cleaned and re-applied. Under this treatment, the painful symptoms of the disease were greatly relieved; and the morbid surface gradually assumed a more healthy appearance." (*Journal of Cutaneous Medicine*, April.)

Tetanus treated with Calabar Bean.—Drs. Boslin and Curron (*Chicago Medical Journal*) have treated a case of acute traumatic tetanus of violent character with large doses of morphia and calabar bean. For a portion of the time, a grain and a half of morphia and three grains of the powdered bean in glycerine were given every hour, with the manifest effect of quieting the patient and relieving the spasm. The patient recovered.

The Therapeutics of Bismuth.—In the *California Medical Gazette*, Dr. W. F. Mac Nutt has a paper on “Some of the Uses of Bismuth,” in which he states that he finds this drug more valuable than it is often supposed to be. Some of his ideas on its therapeutics are novel. “I believe,” he says, “that bismuth not only destroys the sulphuretted hydrogen present in the bowels, but is an antiseptic to albuminous matters, preventing their putrid decomposition. That bismuth destroys the sulphuretted hydrogen present in the bowels, is proved by the fact that if administered for a few hours in considerable quantity the flatulence disappears; and if a dose of oil is given, the evacuations are as black as tar, where the evacuations were natural or clay-coloured before the bismuth was given. It is the chemical action of the gas upon the bismuth which gives the evacuations their black colour. But a small portion of bismuth, when given in powder and in doses from gr. v to gr. xx, is dissolved in the stomach and absorbed. The remainder passes undissolved into the bowels, and while it may have some local anæsthetic action on the bowels as on the stomach, it will be comparatively inert unless there be sulphuretted hydrogen present. Its action on the sulphuretted hydrogen is more particularly demonstrated when given for chronic diarrhœa. Some have attributed to bismuth astringent, tonic, and sedative properties, on account of their success with it in chronic diarrhœa. Others have given it for the same disease without the slightest benefit, and consequently have denied that it has astringent, or sedative, or tonic properties. While the fact is, that in cases of diarrhœa that are caused or kept up by the poisonous effect of sulphuretted hydrogen, I have given bismuth, combined with a few grains of Dover’s powder, with more real benefit to the disease than any drug I could administer. Opium alone is useless, or worse. Charcoal, by absorbing the gas, has been, next to bismuth, the most beneficial. Chambers on ‘Indigestions’ says that it is ‘rare to find sulphuretted hydrogen or hydro-sulphate of ammonia excreted without watery or soft pultaceous stools. They appear to be purgative poisons.' They are not always purgative poisons."

Physiological Action of Absinthe.—The following are the conclusions which M. Magnan laid before the French Academy of Sciences, and which have given rise to some discussion in the weekly journals:—1. The epileptic or epileptiform accidents in alcoholism—or, in other words, alcoholic epilepsy—are of a radically different nature, according as the alcoholism is acute or chronic. 2. In acute alcoholism the epilepsy is under the complete influence of an external agent, of a poison (absinthe) which of itself alone causes the epileptic attack; it is epilepsy by "intoxication." 3. The alcoholic epileptics exhibit the ordinary features of simple alcoholic cases, and also superadded phenomena, among which the epileptic attack is dominant. 4. These two groups of symptoms (the alcoholic symptoms and alcoholic convulsions), united in the same subject, have a relation to the twofold nature of the poison (absinthe), whose elements are absinthe and alcohol. 5. In chronic alcoholism the epileptic or epileptiform accidents are under the direct control of organic modifications which take place in the patient. The excess of liquids, in gradually altering the tissues, renders them capable, under the influence of various causes, of producing by themselves convulsive epileptiform phenomena, accidents analogous to those that we see take place in other patients in certain cases of lesions of the nervous centres (general paralysis, tumours of the brain, &c.) (*Comptes Rendus*, April 5.)

Notes and Queries.[13]

The Dose of Atropia for Subcutaneous Injection.—I wish to answer the query of Dr. Sisson in the *Practitioner* of last month with every caution, but I have formed a strong opinion on the subject in question. A "Country Practitioner," who wrote to this journal in February, states that he used to give hypodermic injections of more than ⅕ grain of atropia daily, to the same patient, for years. Now I stated in my paper in this journal (July 1868), that such doses as these are utterly unsafe, and I retain that opinion. Supposing that the sulphate of atropia was good, which it is very possible it was not in the "Country Practitioner's" case, I can affirm, from my own knowledge, that there are many patients to whom such a dose would be dangerous and probably fatal, if it were really fairly introduced into the subcutaneous tissue. *I have seen uncomfortable atropism from the injection of less than $\frac{1}{100}$ grain*: a case occurred to me only a week or two since. It is therefore unadvisable to begin, at any rate, with large doses. With a large experience of subcutaneous injection, I am enabled to say with confidence that $\frac{1}{60}$ or $\frac{1}{50}$ grain doses are what are best borne by the majority of persons; that sometimes, but not often, it is necessary to go as far as $\frac{1}{30}$ grain; and that not unfrequently patients will not bear as much as the $\frac{1}{60}$ without uncomfortable atropism. Hence my recommendation to practitioners to commence, experimentally, with such a small quantity as the $\frac{1}{120}$ grain.—F. E. Anstie.

Strychnia as a Remedy in a Severe Case of Nervous Hyperæsthesia.—Mr. H. A. Allbutt, of Leeds, writes to us:—"The following case may be of some interest to the readers of the *Practitioner*, as in some of its symptoms it presented some curious phenomena. Mrs. A——, a married lady, about thirty-six years of age, consulted me in last November for lameness, and great pain and difficulty in walking, with obscure pains in her back and sides, severe palpitation, restless nights, loss of appetite, and great nervousness. She often, too, complained of dimness of vision, and the thumb was at times flexed across the palm of the hand in a spasmodic manner. In addition to these symptoms, she suffered from prolapsus uteri and menorrhagia. The difficulty and pain in walking were, however, the most prominent features of the case. She could not raise her feet the height of a step, and her locomotion was most curious, consisting of a sort of corkscrew motion, or twisting of the foot and thigh each step that was taken. This had been going on for three years, sometimes better and sometimes worse, but on the whole she was gradually getting worse. I learned from her that when younger she had been of a strong, healthy nature, and suffered from little illness till her marriage. She seems to have suffered severely during her various confinements from floodings, &c., and from the time of her last confinement she has been affected more or less in the manner described. In regard to the cause of this condition, I am of opinion that her nervous system had been much weakened, and thrown into an excitable condition by the shocks of labour and by the floodings at those times. In this opinion I was borne out by Dr. Allbutt, of Leeds, who saw the case with me. The treatment at first consisted of the various preparations of steel, of which I found the ferri ammon. cit., combined with bromide of potassium, to be the best. Tannic acid pessaries were also ordered to be introduced into the vagina for the relief of the prolapsus uteri, and a hypodermic injection of morphia was given every night for three weeks. Under the influence of these remedies she improved in her general health, and the lameness was improved in a slight degree; in fact, she seemed to arrive at a certain standard of health and to advance no further. I was now induced to try strychnia, in the form of pills, combined with carbonate of iron. The dose was $\frac{1}{20}$ gr. twice a day. The effects were marvellous. The lameness is fast disappearing, and she is able to walk out of doors, which she has not done for fifteen months. I firmly believe she will be quite cured if the treatment is persevered in.”

Muriate of Ammonia as a Remedy.—Dr. Cholmeley has kindly sent us a note, the manuscript of which, by an unfortunate accident, has been mislaid. The substance of his remarks is as follows:—He confirms the observations of Dr. Anstie, in a paper in the December number of the *Practitioner*, as to the great efficacy of the muriate of ammonia as a remedy for neuralgic and myalgic pain. But Dr. Cholmeley goes on to say, that with regard to a matter on which Dr. Anstie spoke more doubtfully,—the efficacy, namely, which certain authors have ascribed to this drug as an emmenagogue,—he has formed from a large experience a decided opinion in favour of the utility of this medicine. He is convinced that in a very large number of cases of absent or suppressed menstruation, muriate of ammonia acts in a very direct and powerful manner in establishing or restoring the flux. Dr. Cholmeley has now experimented with the muriate, in doses of 10 to 20 grains, in so large a number of hospital and dispensary patients, that he cannot suppose there is any room for fallacy in this conclusion.

[Since the date at which the paper referred to by Dr. Cholmeley was written, we have had occasion to employ the muriate of ammonia in two cases of amenorrhœa, with apparently very striking and direct results of a curative kind. As yet, however, we must confess ourselves unable to lay down any definite rule as to the class of cases to which it is applicable with the best chance of success, beyond a general idea that it acts best in persons not anæmic, but possessing a weak and mobile nervous system.—Eds. Pract.]

Ether Spray in Operations about the Anus.—Dr. John Barclay, of Banff, writes to us as follows:—“I write to corroborate what was written by Mr. Alexander Bruce, in the *Practitioner* for last month, concerning the employment of ether spray in operations about the verge of the anus. I have experience of it in two cases. The first was the slitting up of a hæmorrhoid containing a clot, and when the ether spray was directed on the part the patient screamed in intense agony, comparing it to nothing else than the introduction of a red-hot iron. The pile was opened without it, and the patient said the cutting was as nothing compared with the spray. The second case was very similar to this, and the result here was the same. So that I never dream now of recommending the freezing by ether in operations in that region. I may remark that it is a curious thing ice never seems to give pain when so employed.”

Treatment of Hæmorrhoids.—Mr. J. Christophers, of Wadebridge, Cornwall, sends us the following note:—“The pain and risk attending operations for the removal of hæmorrhoids, whether by knife, ligature, cautery, or caustic, render valuable any less heroic mode of treatment, whereby the necessity for using the means alluded to may be dispensed with, or even rendered less frequent. The term hæmorrhoid or pile being used to signify a tumour caused by enlarged or varicose veins at the lower part of the rectum, the definition of the disease would seem to indicate its treatment—pressure and support. The benefit resulting from pressure on tumours, and from pressure and support applied to varicose veins situated on the surface of the body, is manifold and manifest. The same good results often attended pressure internally applied in cases of hæmorrhoids, and frequently in cases of prolapsus also. Occasionally after having introduced the finger into the rectum, in cases of hæmorrhoids, for the purpose of exploration, I have heard with surprise the patient affirm that the examination had temporarily relieved the severity of his pain. Continuous pressure exercised by means of the rectum plug, of a size, form, and material suited to these cases, in many instances, affords immediate relief, and often effects ultimate cure; the rectum plug being nothing more than a simple cone or peg, terminating in a short stem or disk, having a hole bored through its long diameter formed of metal, ivory, wood, membrane, or of other material capable of inflation. Any of these substances answer the purpose, some being suited to one kind of case, some to another. Those formed of wood have in my hands answered well, and have often achieved a success that has exceeded my expectations. The shape and size best suited to individual cases experience soon teaches. A not unfrequent obstacle in treating cases by the rectum plug will be found to consist in the intolerance by the rectum in some patients at first of its presence; perseverance in its use gradually and surely overcomes this difficulty. After a short probation all discomfort ceases, and the plug can be worn by day and by night, sitting, riding, walking, or standing, with the best results, and that not only in cases of hæmorrhoids, but in bad cases of prolapsus also. So much is this the case that many who have worn a rectum plug, though with difficulty at the beginning, give up its use, even when the malady that demanded its application is cured, with reluctance and regret, saying that they derive comfort and support from its presence. These circumstances induce me to think that this safe and simple means of treating hæmorrhoids has been too much neglected, and for this reason I venture to bring it under the notice of the *Practitioner*."

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1. *Boston Medical and Surgical Journal*, May 21, 1868.

2. "The Restorative Treatment of Pneumonia." Third Edition. Edinburgh: Black, 1866.

3. *British Medical Journal*, December 28, 1867.

4. *British Medical Journal*, February 22, 1868.

5. See also some Lectures on Pneumonia by Dr. Waters, of Liverpool (*British Medical Journal*, October 1867), whose views and treatment, allied to those of Dr. Sieveking—I hope he will excuse me for thinking—are very unsatisfactory, when compared with the results obtained by a restorative practice.

6. *Practitioner,* November 1868.

7. Read at the third Quarterly Meeting of the Medico-Psychological Association, held at the Royal Medico-Chirurgical Society, April 29, 1869.

8. Extract from Nineteenth Annual Report of the Somerset Asylum:—"One female maniac, C. L., aged 35, single, most obscene in her conduct and language, noisy, destructive, and dirty, got rapidly well after the employment of the hypodermic injection of a solution containing half a grain of acetate of morphia."

Extract from Twentieth Annual Report of the Somerset Asylum:—"The hypodermic injection of about half a grain of acetate of morphia in ♏x. of distilled water has been useful in cases of maniacal excitement with sleeplessness."

9. The gas used in the following instances was supplied to me by Barth, of London, and administered in his apparatus. I would wish, in this place, to thank Dr. Birch, of Kensington, for introducing it to my notice, and for his kind communications on the subject of this paper.

10. In estimating the value of oxygen in these cases of phthisis we must bear in mind the mechanical effect of deep and steady inspiration through a long tube; this, *per se*, has a tendency to expand the lung vesicles and to hasten the healing of cavities, as has been fairly shown by Ramadge, in spite of his absurdities.

11. Cf. Birch on "Action, &c. of Oxygen," 2d edit. p. 33.

12. Our own experience is very favourable to petroleum.—EDS. PRACT.

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